

BFS Traversal

Step 1: Start at E

Legend

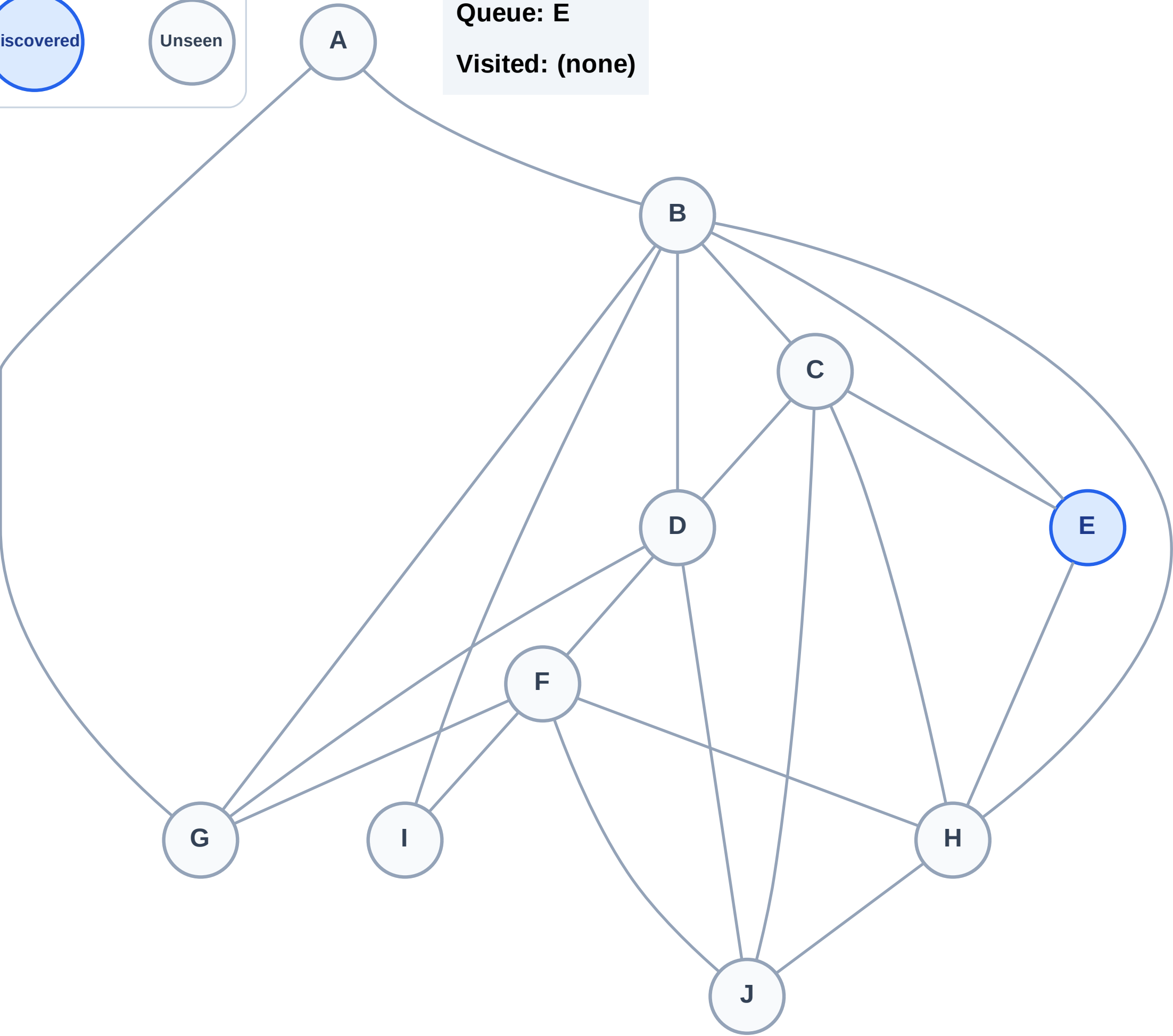
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

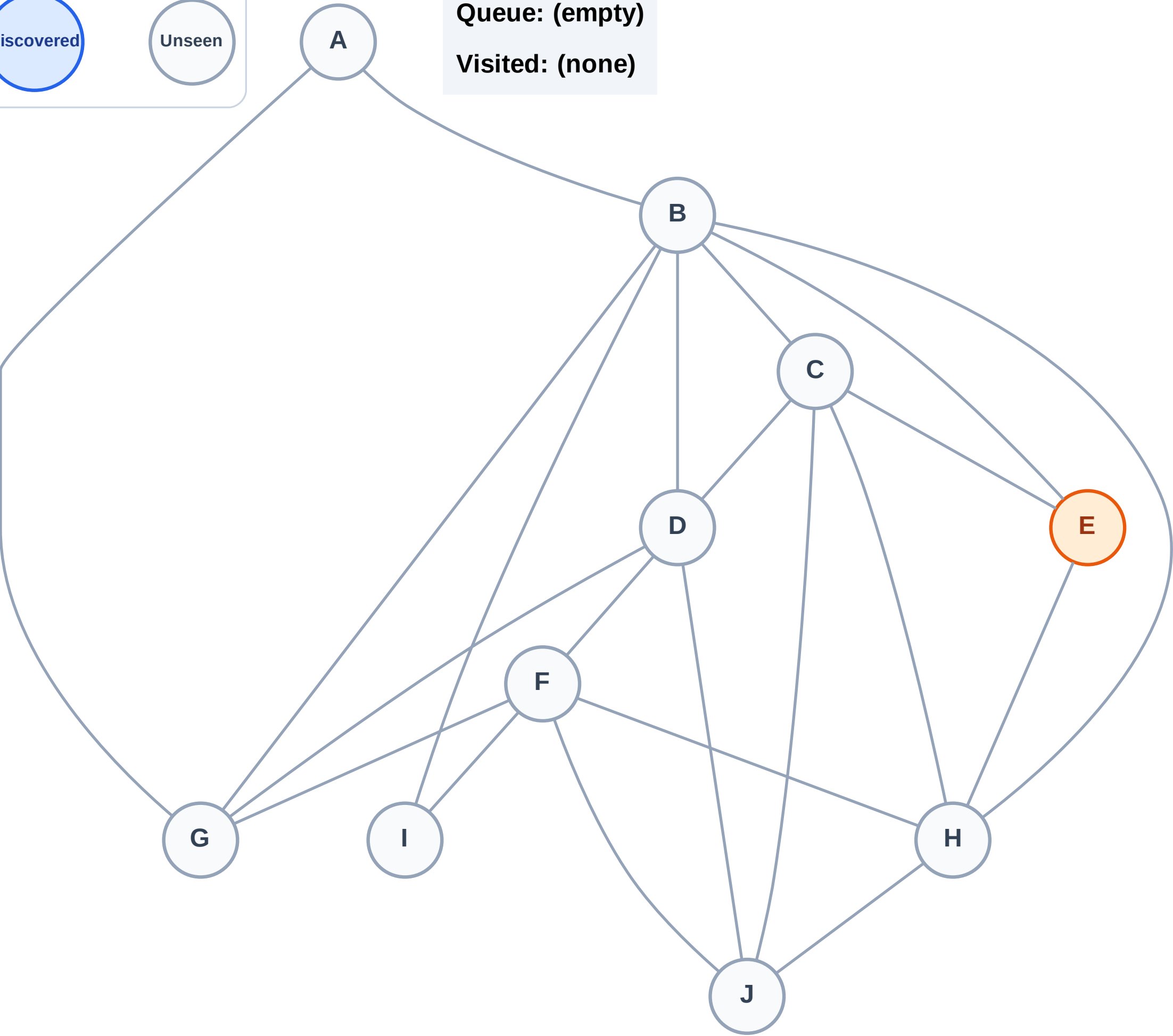
Complete

Processing

Discovered

Unseen

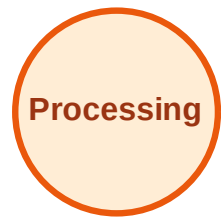
Queue: (empty)
Visited: (none)



BFS Traversal

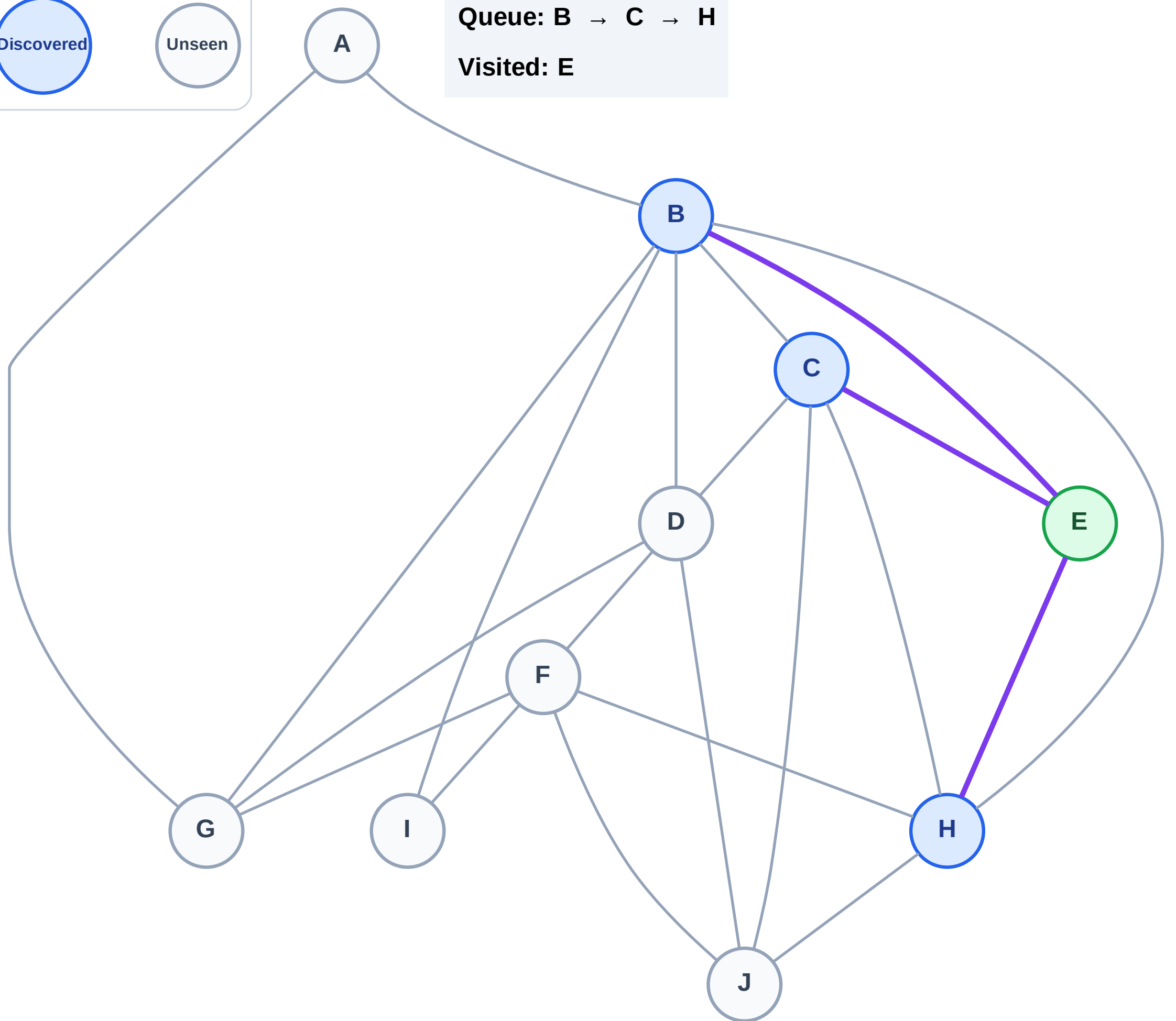
Step 3: Discover B, C, H from E

Legend



Queue: B → C → H

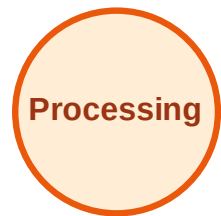
Visited: E



BFS Traversal

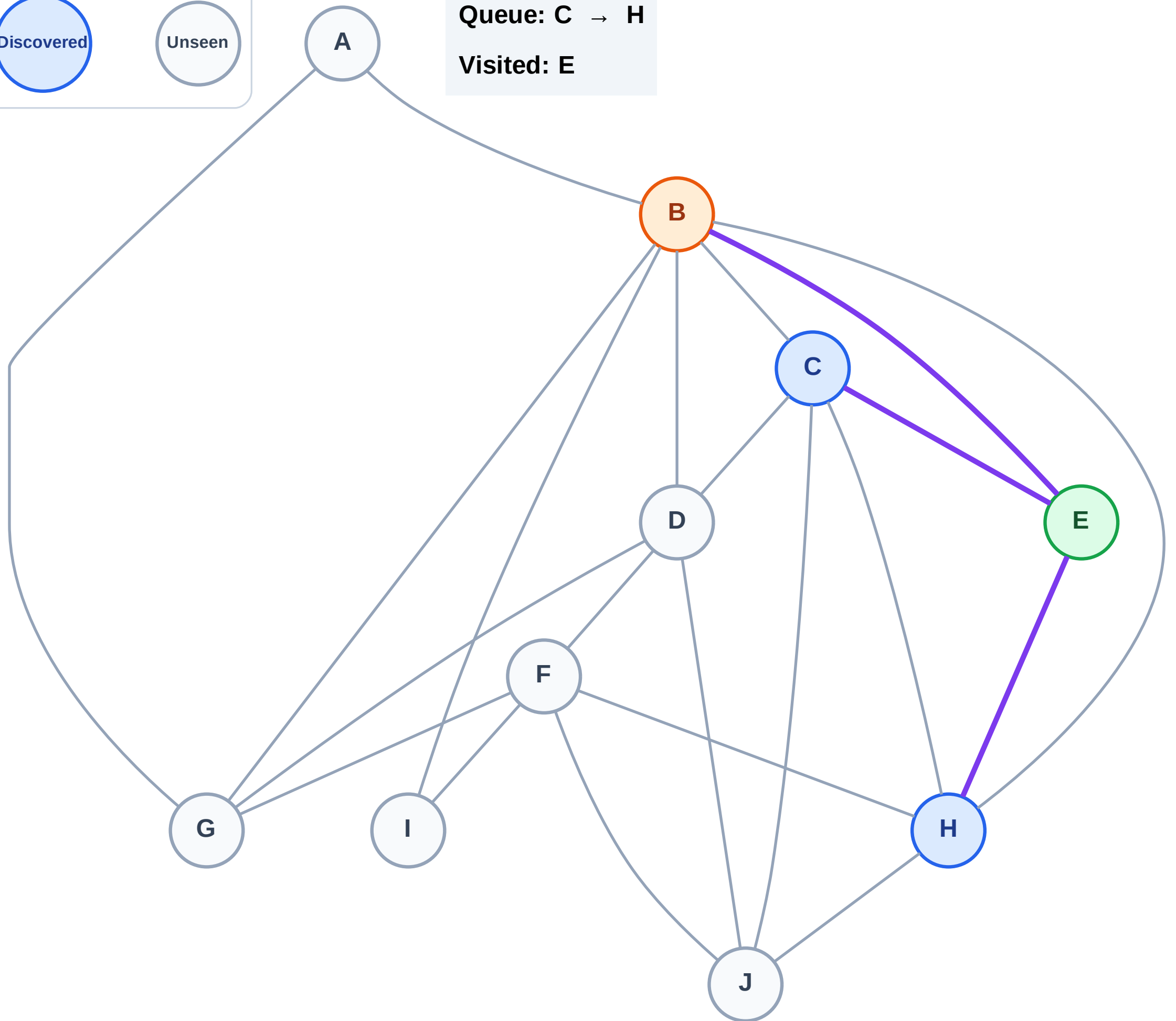
Step 4: Process node B

Legend



Queue: C → H

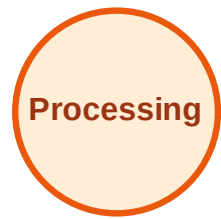
Visited: E



BFS Traversal

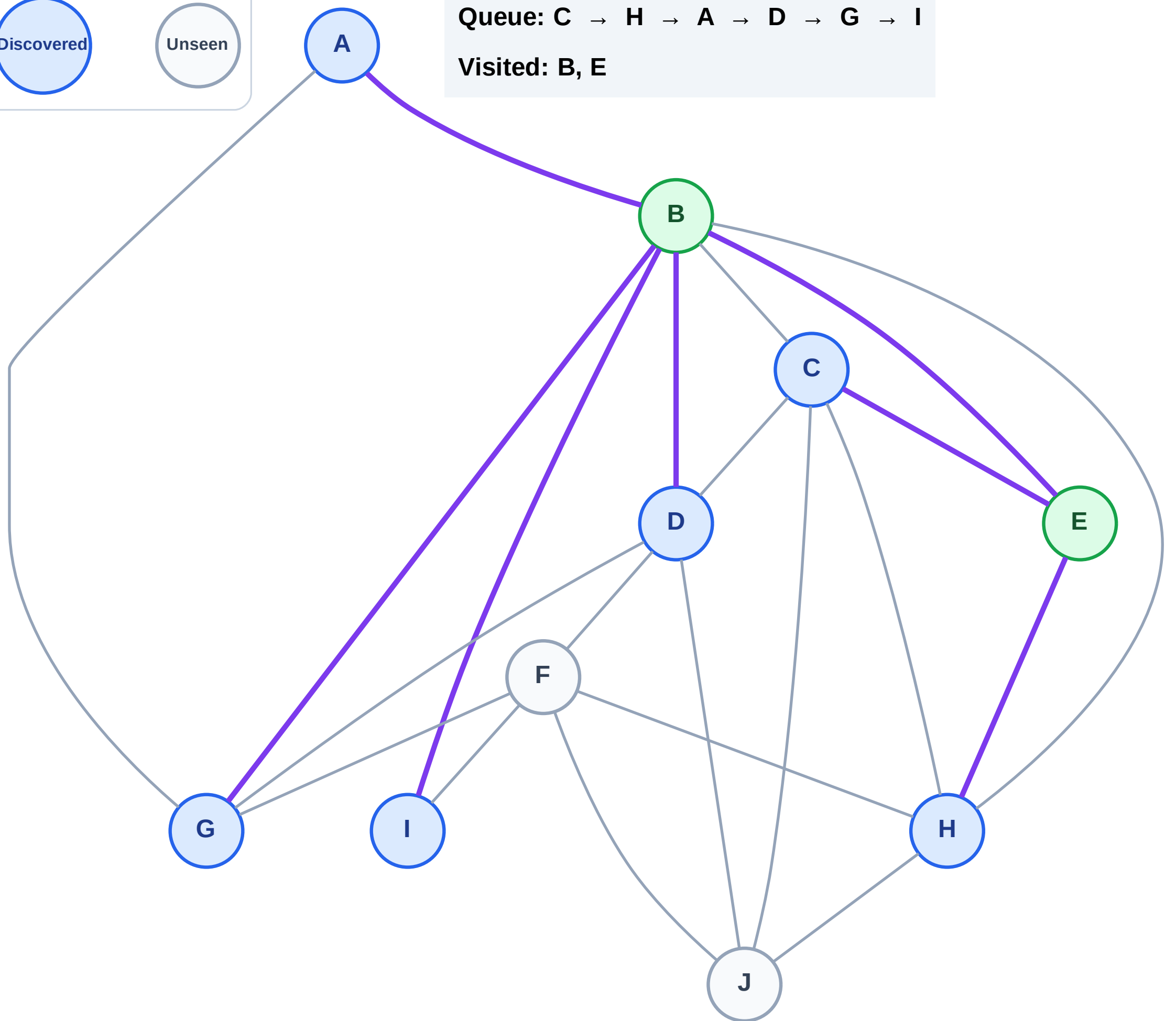
Step 5: Discover A, D, G, I from B

Legend



Queue: C → H → A → D → G → I

Visited: B, E



BFS Traversal

Step 6: Process node C

Legend

Complete

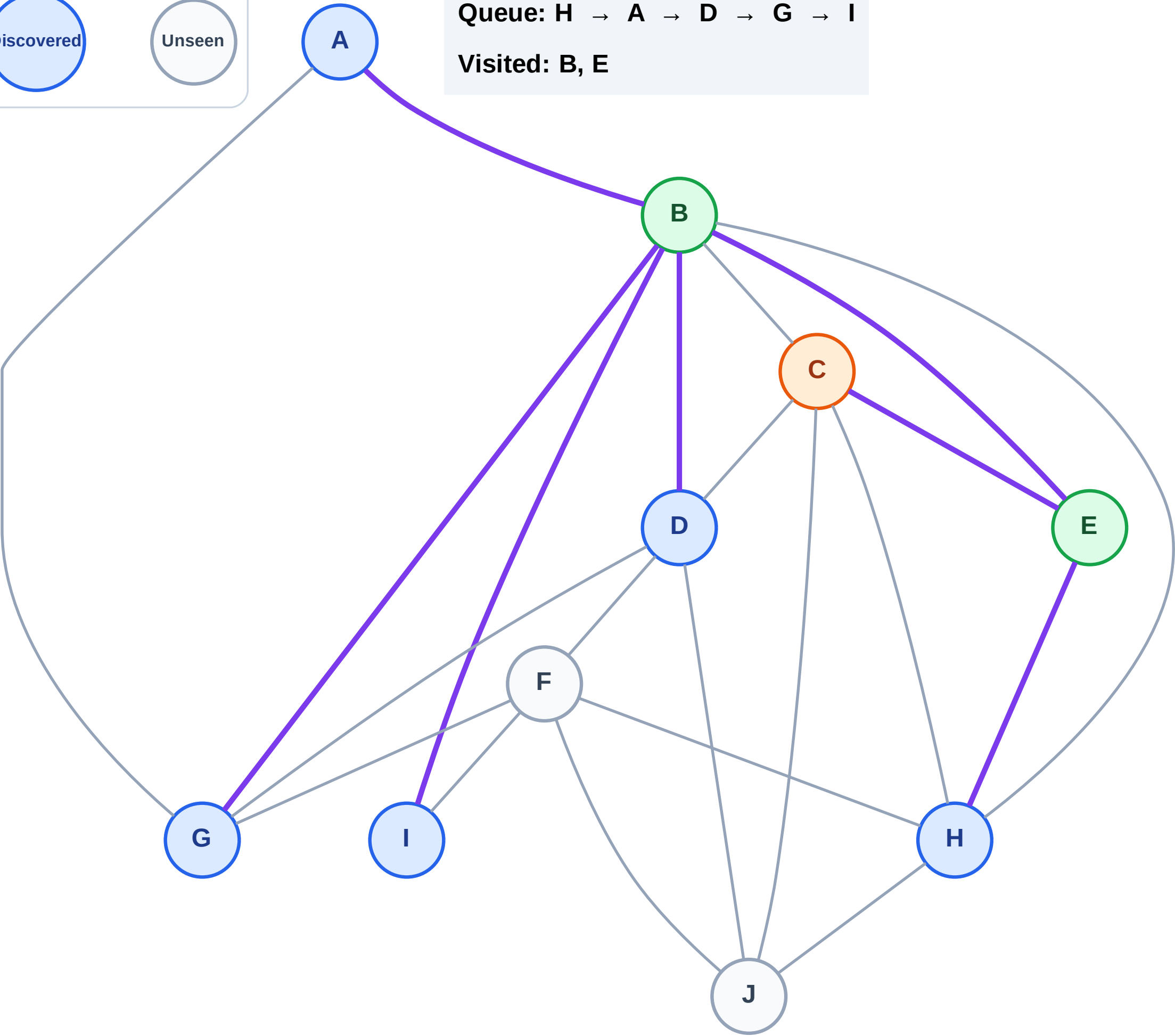
Processing

Discovered

Unseen

Queue: H → A → D → G → I

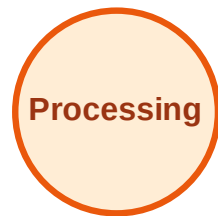
Visited: B, E



BFS Traversal

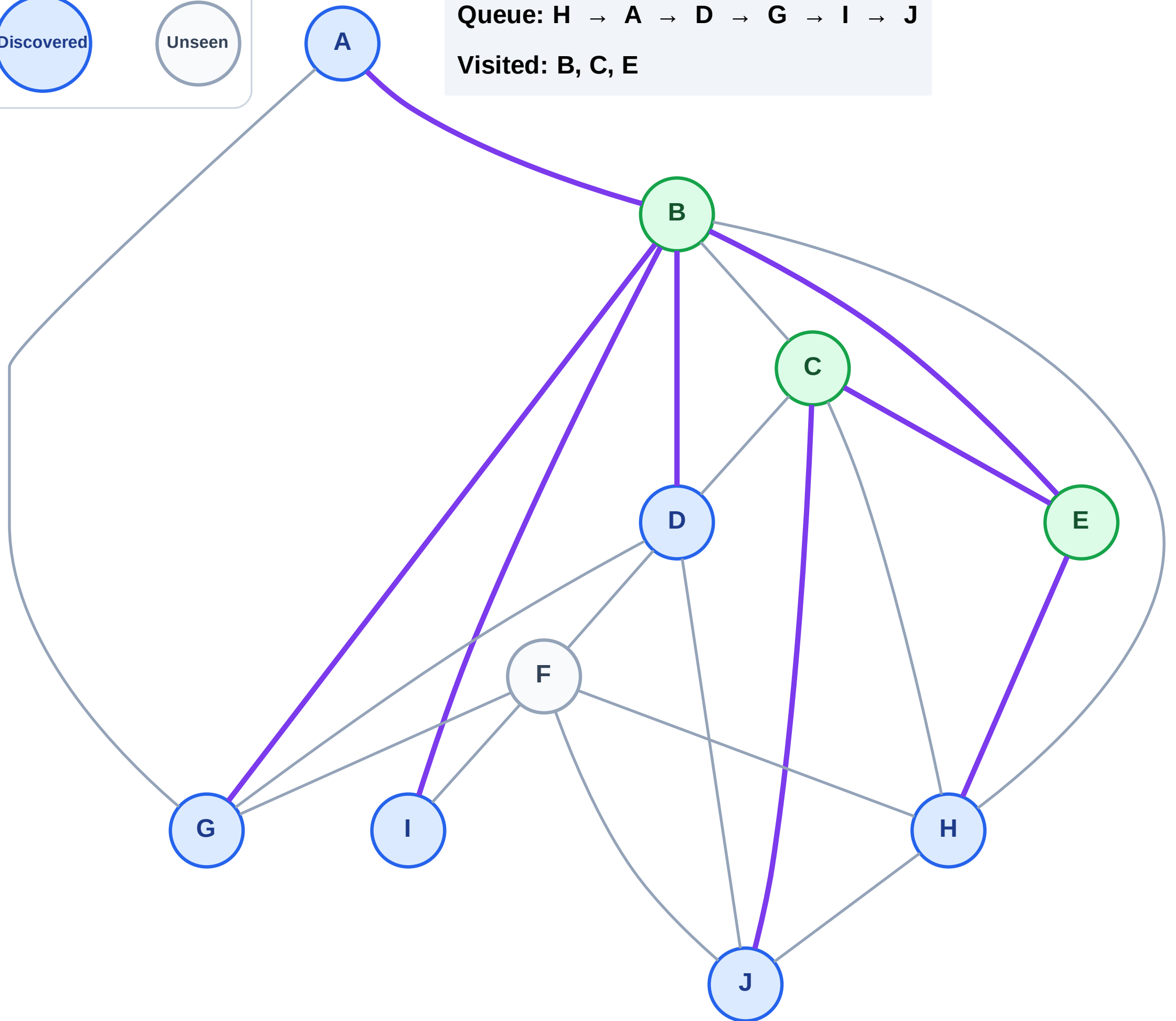
Step 7: Discover J from C

Legend



Queue: H → A → D → G → I → J

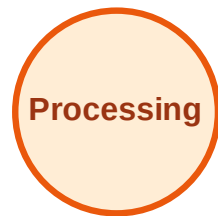
Visited: B, C, E



BFS Traversal

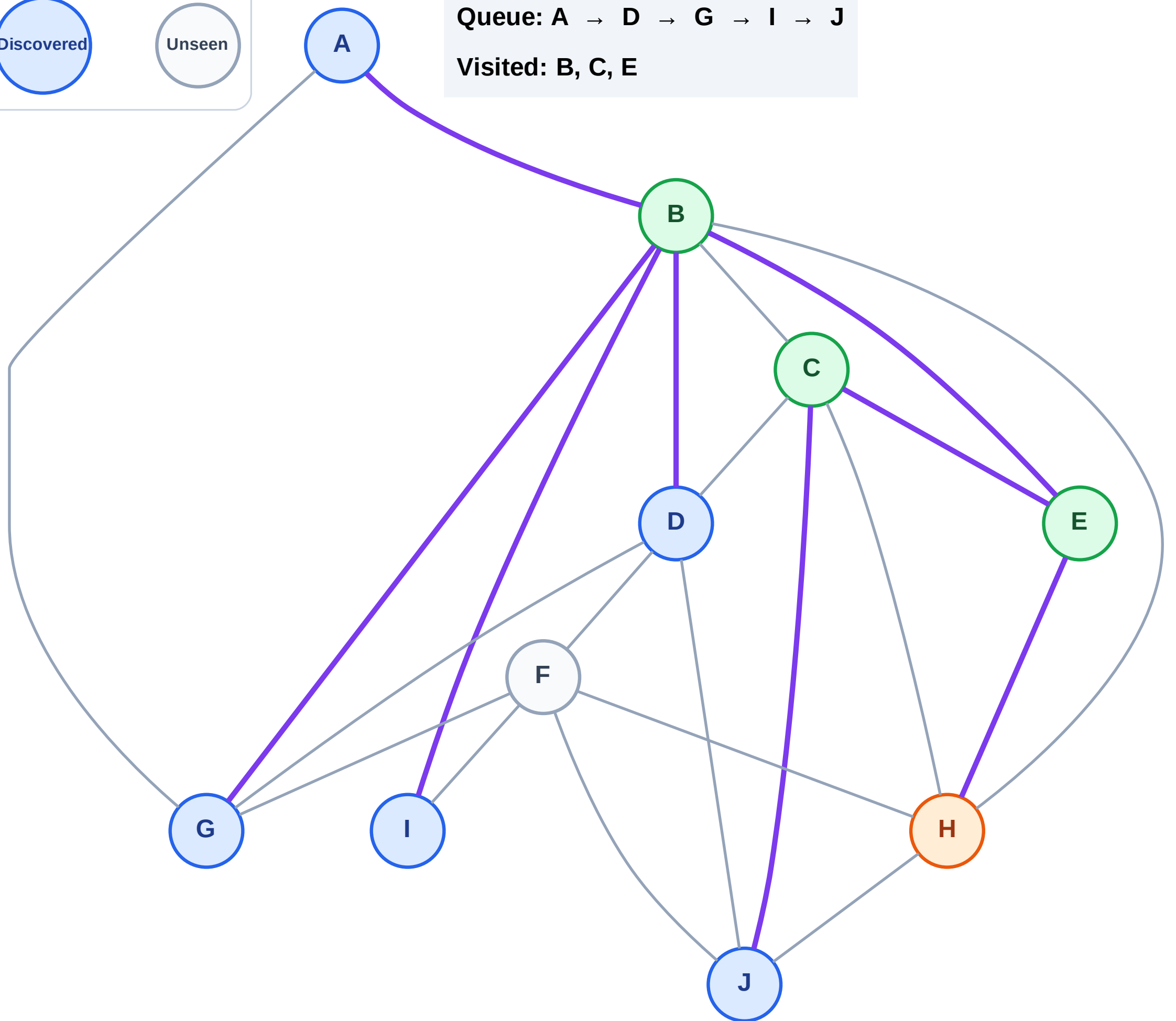
Step 8: Process node H

Legend



Queue: A → D → G → I → J

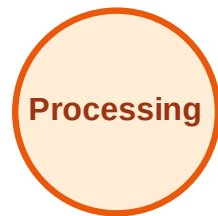
Visited: B, C, E



BFS Traversal

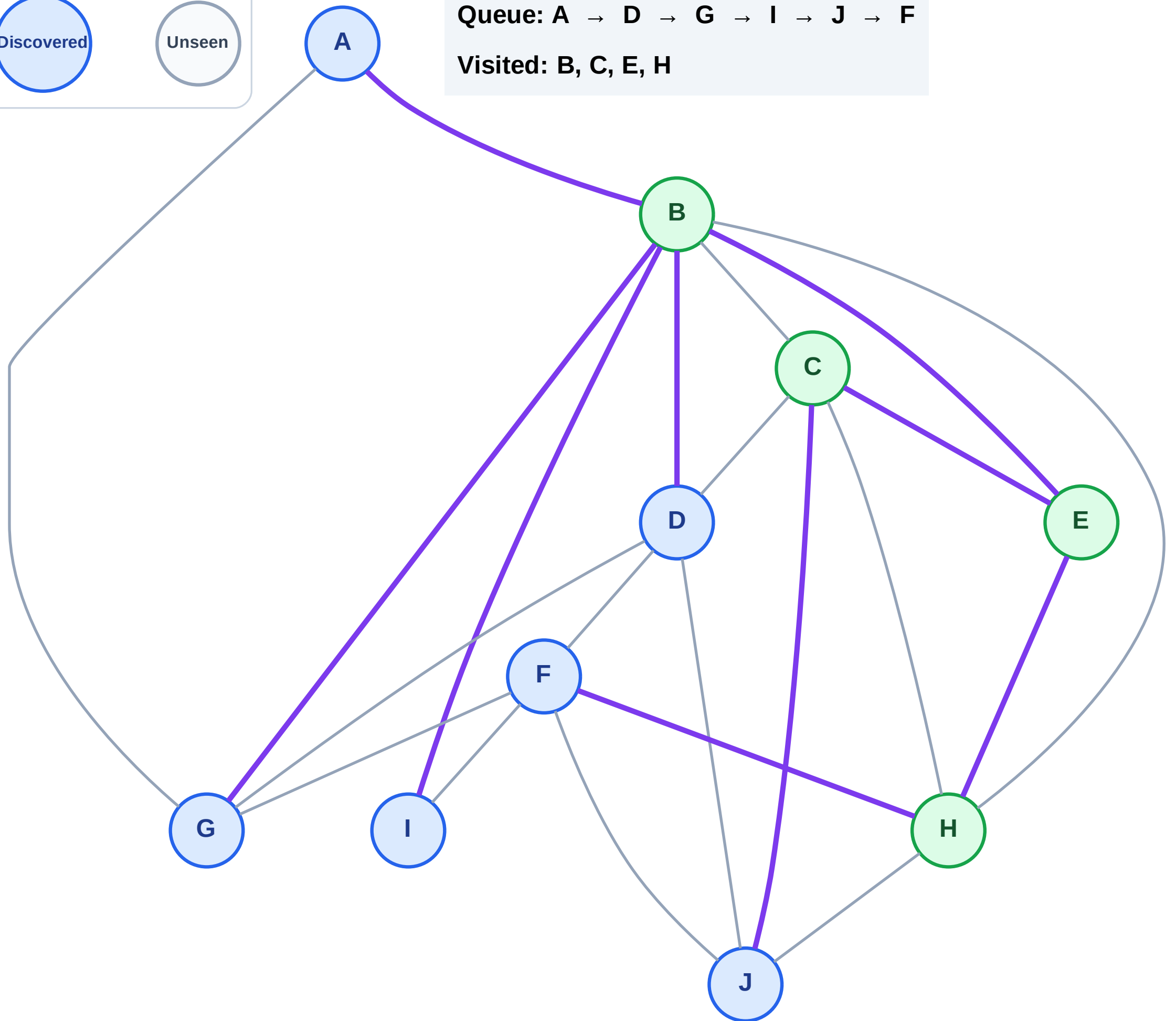
Step 9: Discover F from H

Legend



Queue: A → D → G → I → J → F

Visited: B, C, E, H



BFS Traversal

Step 10: Process node A

Legend

Complete

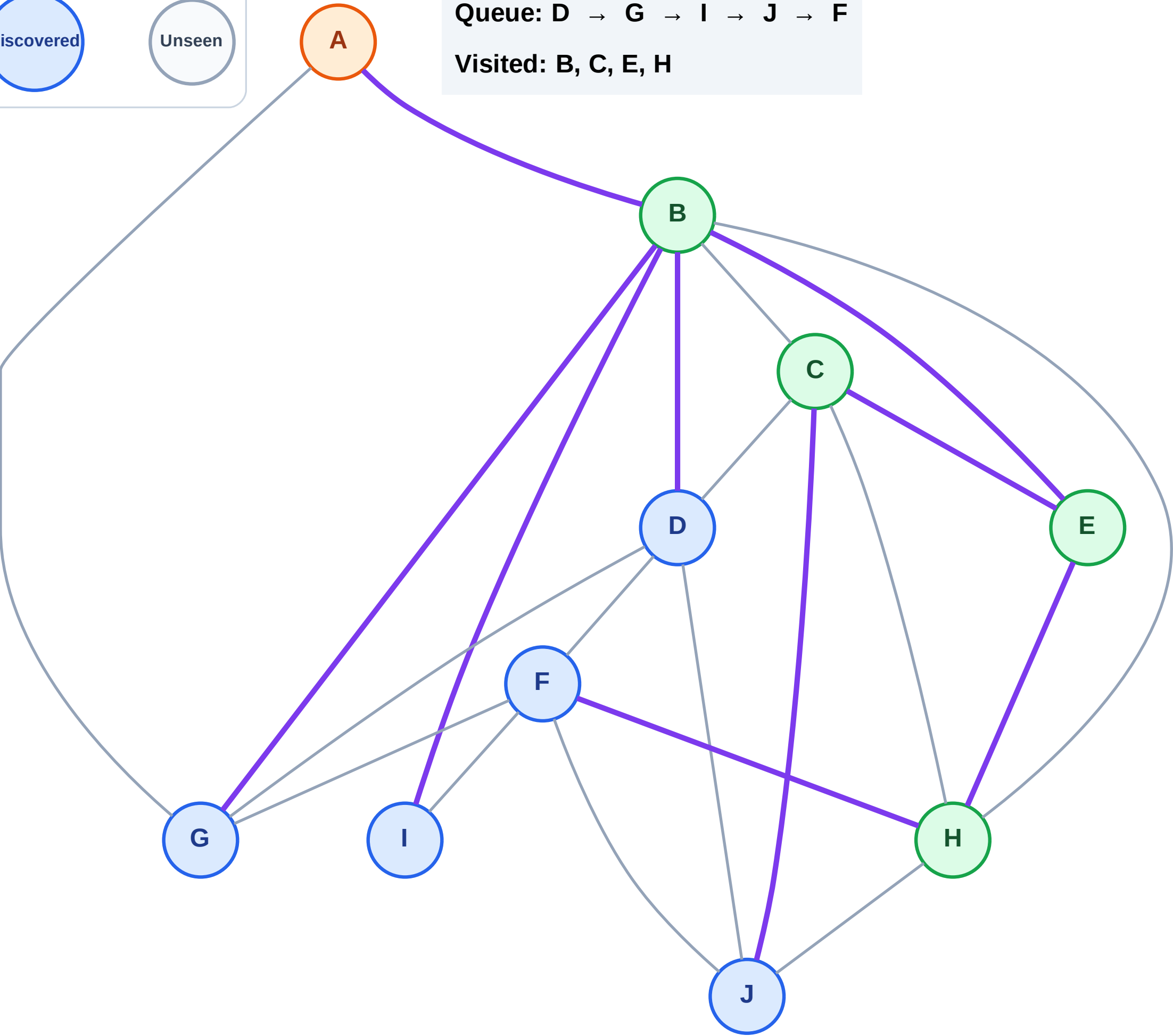
Processing

Discovered

Unseen

Queue: D → G → I → J → F

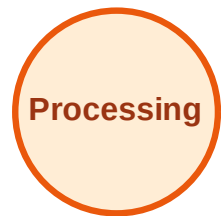
Visited: B, C, E, H



BFS Traversal

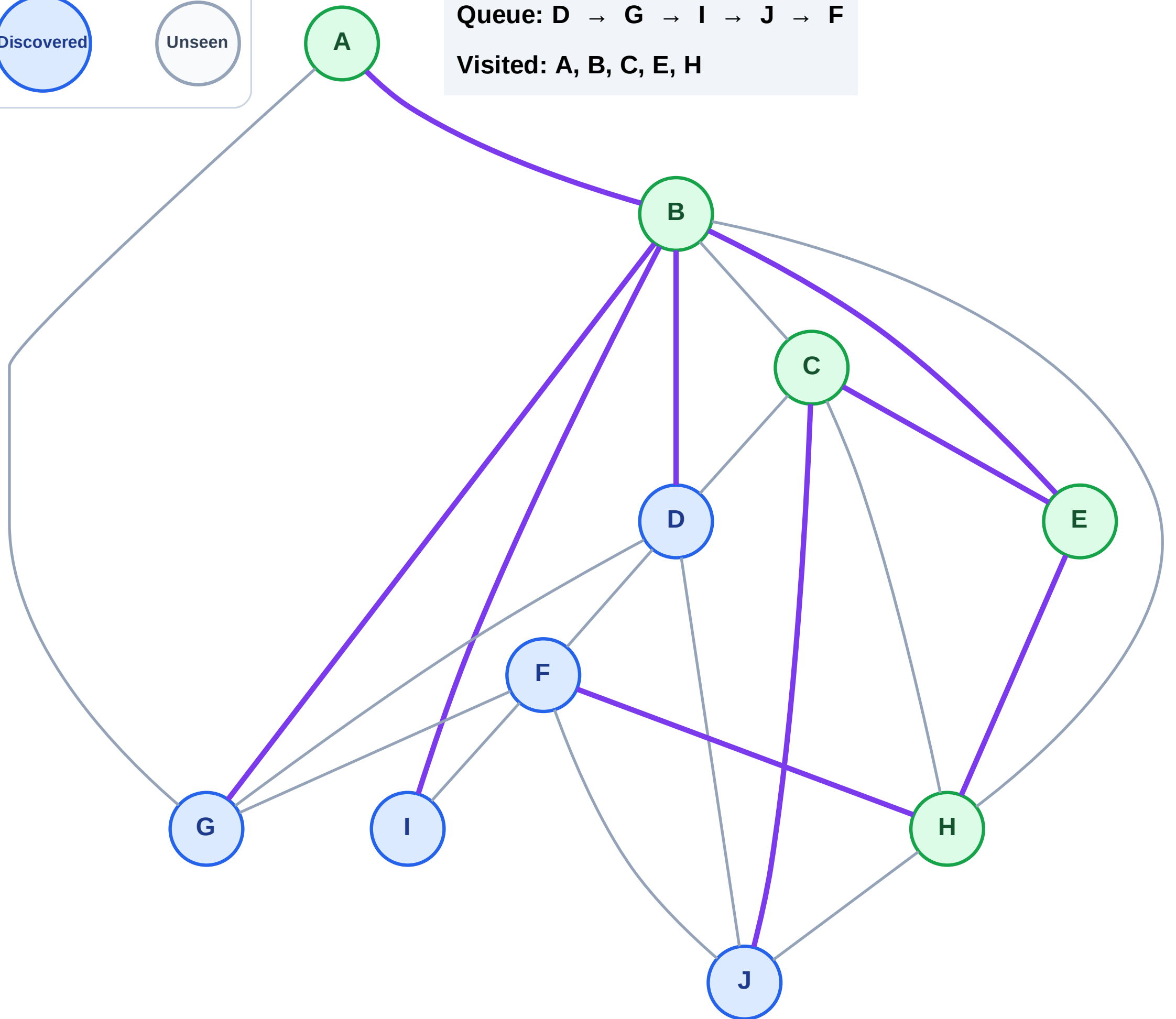
Step 11: No new neighbors from A

Legend



Queue: D → G → I → J → F

Visited: A, B, C, E, H



BFS Traversal

Step 12: Process node D

Legend

Complete

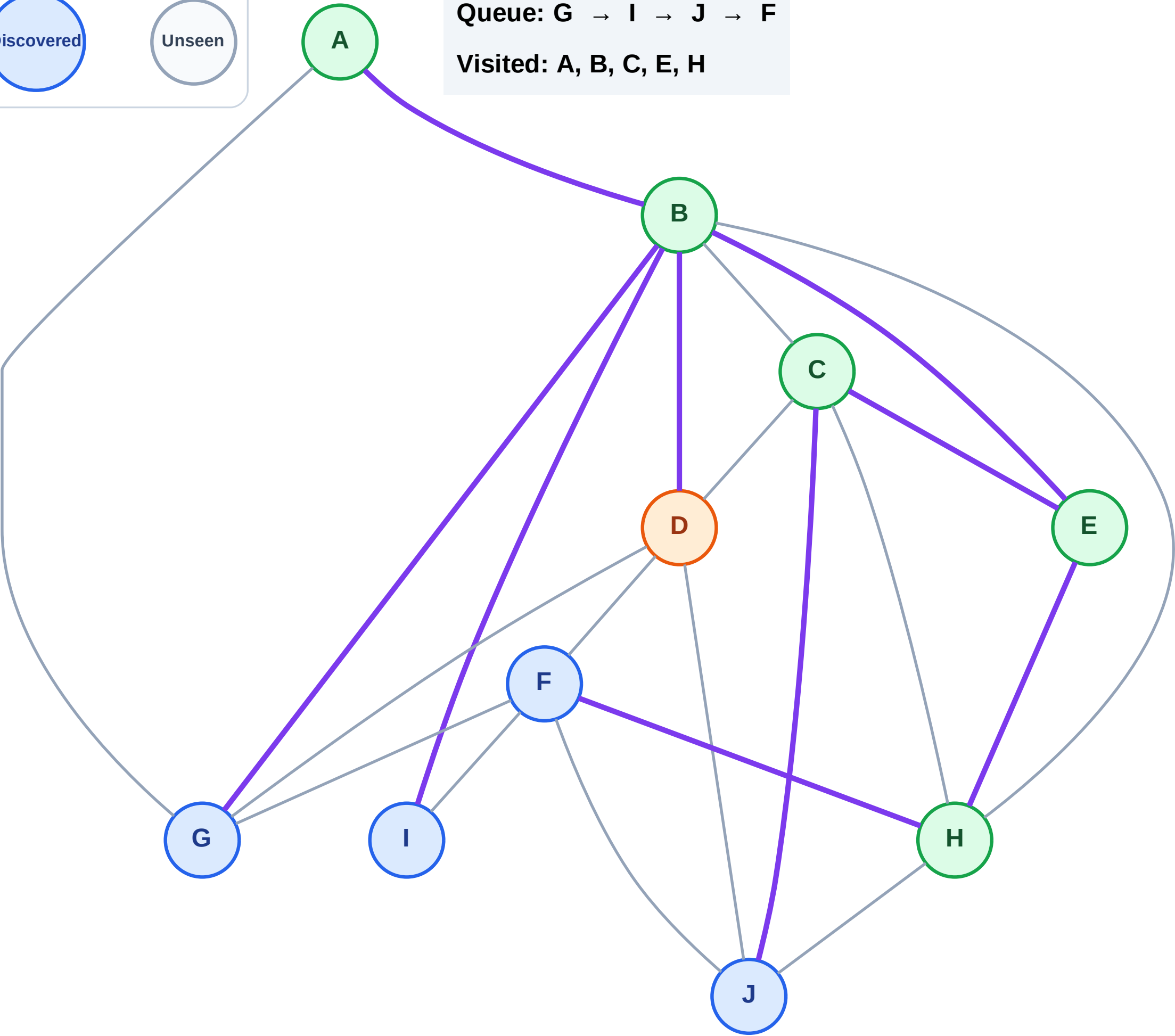
Processing

Discovered

Unseen

Queue: G → I → J → F

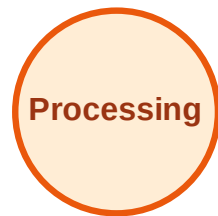
Visited: A, B, C, E, H



BFS Traversal

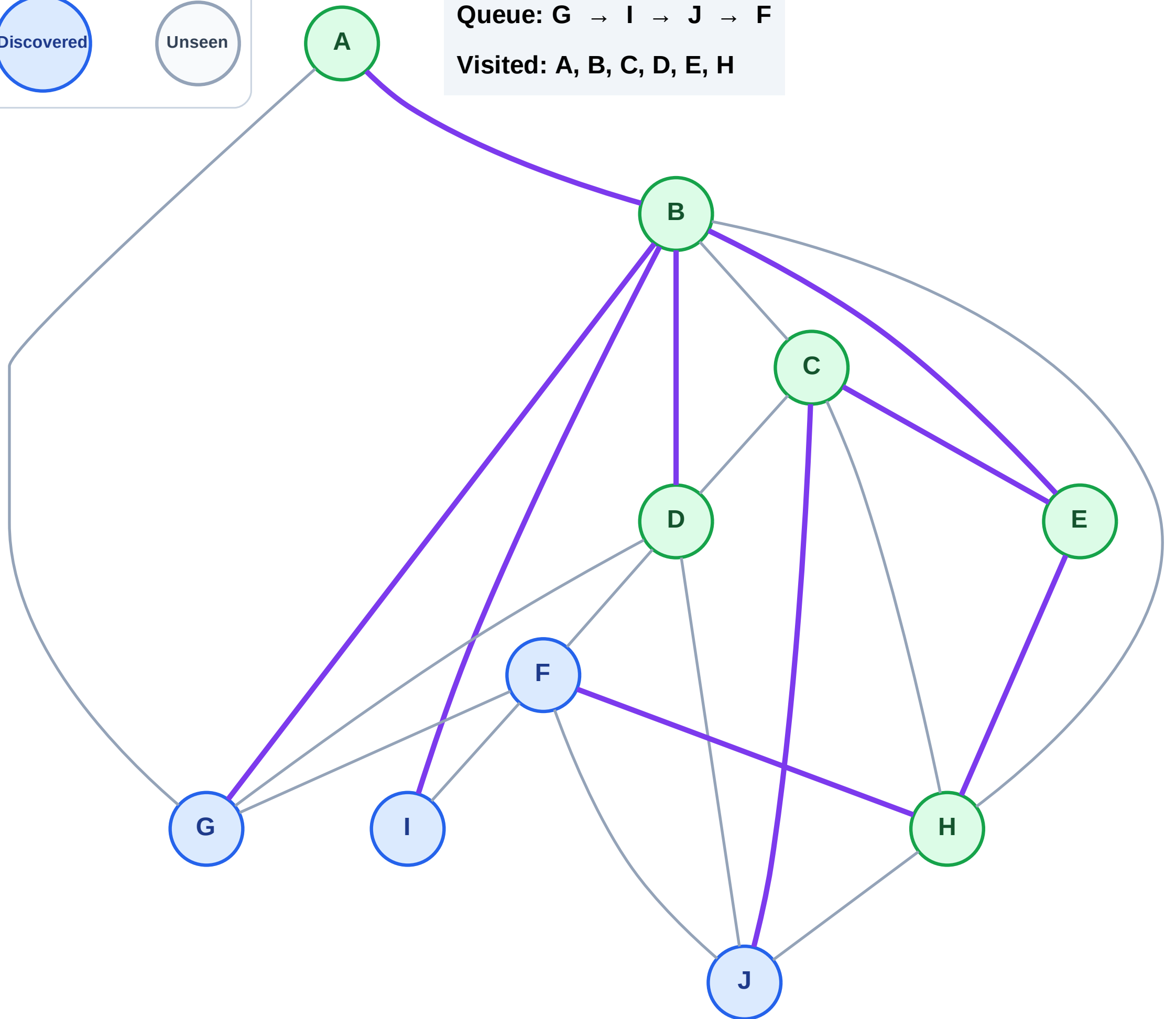
Step 13: No new neighbors from D

Legend



Queue: G → I → J → F

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node G

Legend

Complete

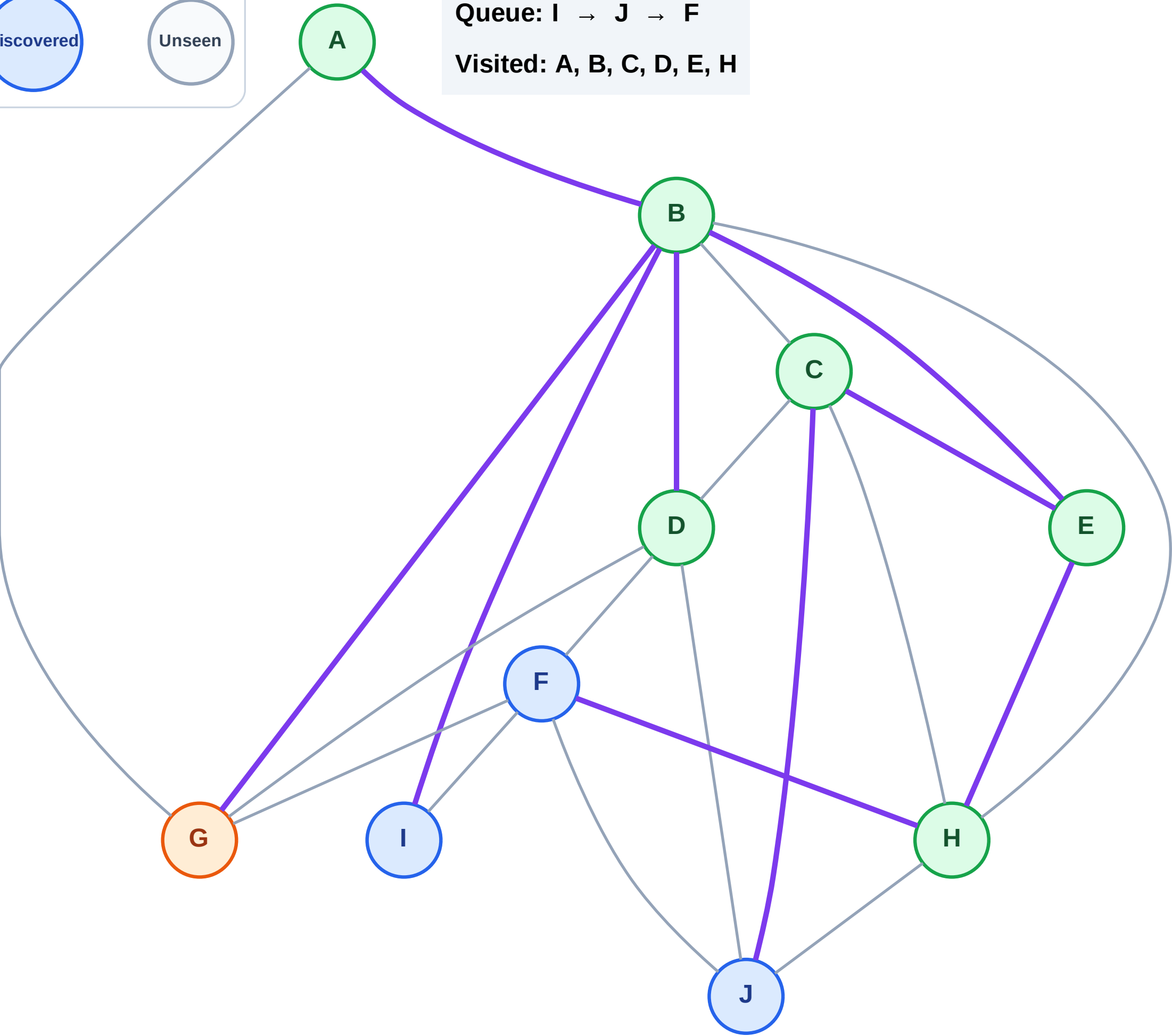
Processing

Discovered

Unseen

Queue: I → J → F

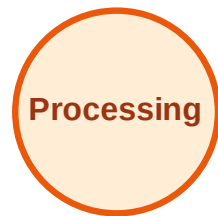
Visited: A, B, C, D, E, H



BFS Traversal

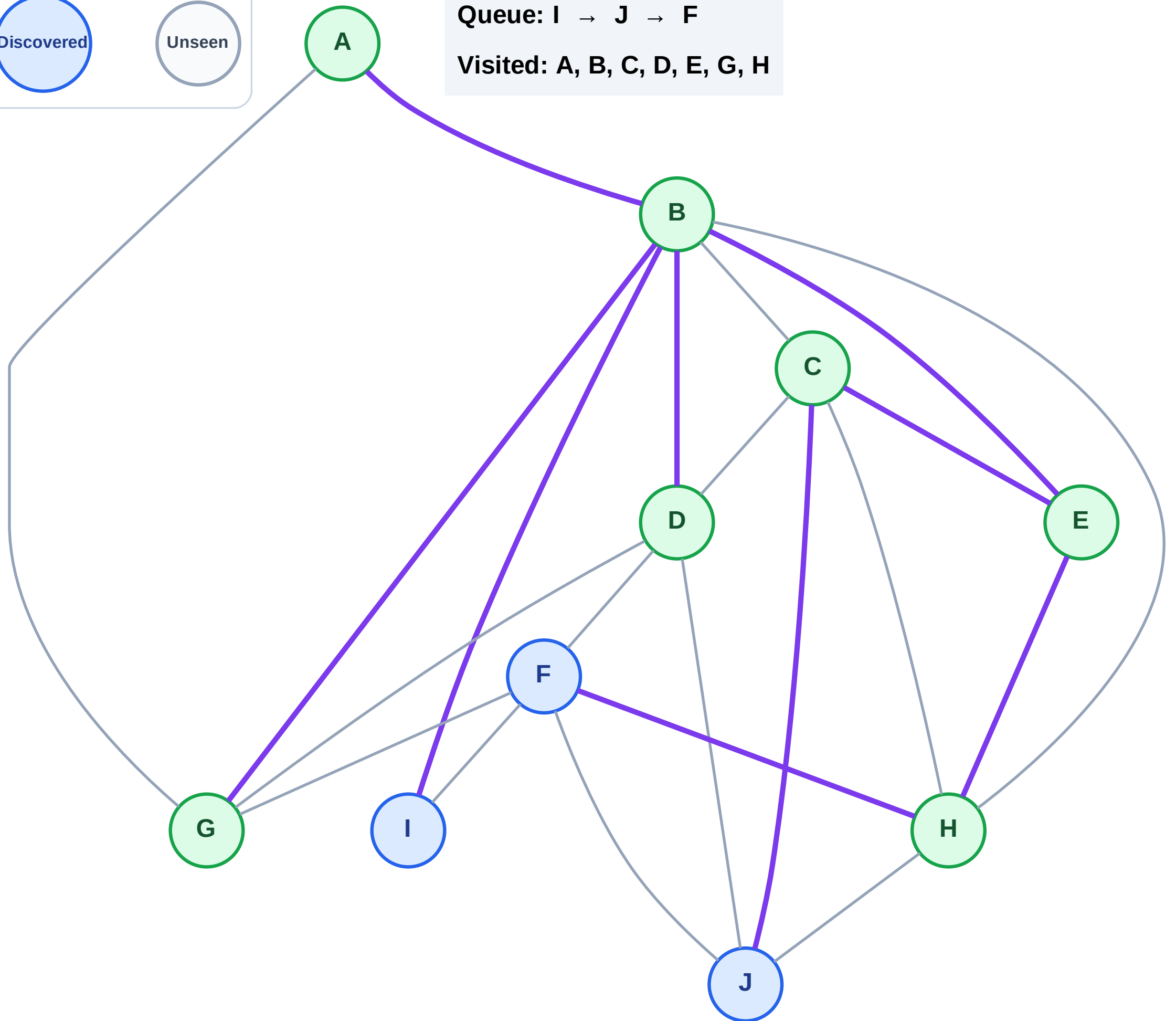
Step 15: No new neighbors from G

Legend



Queue: I → J → F

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node I

Legend

Complete

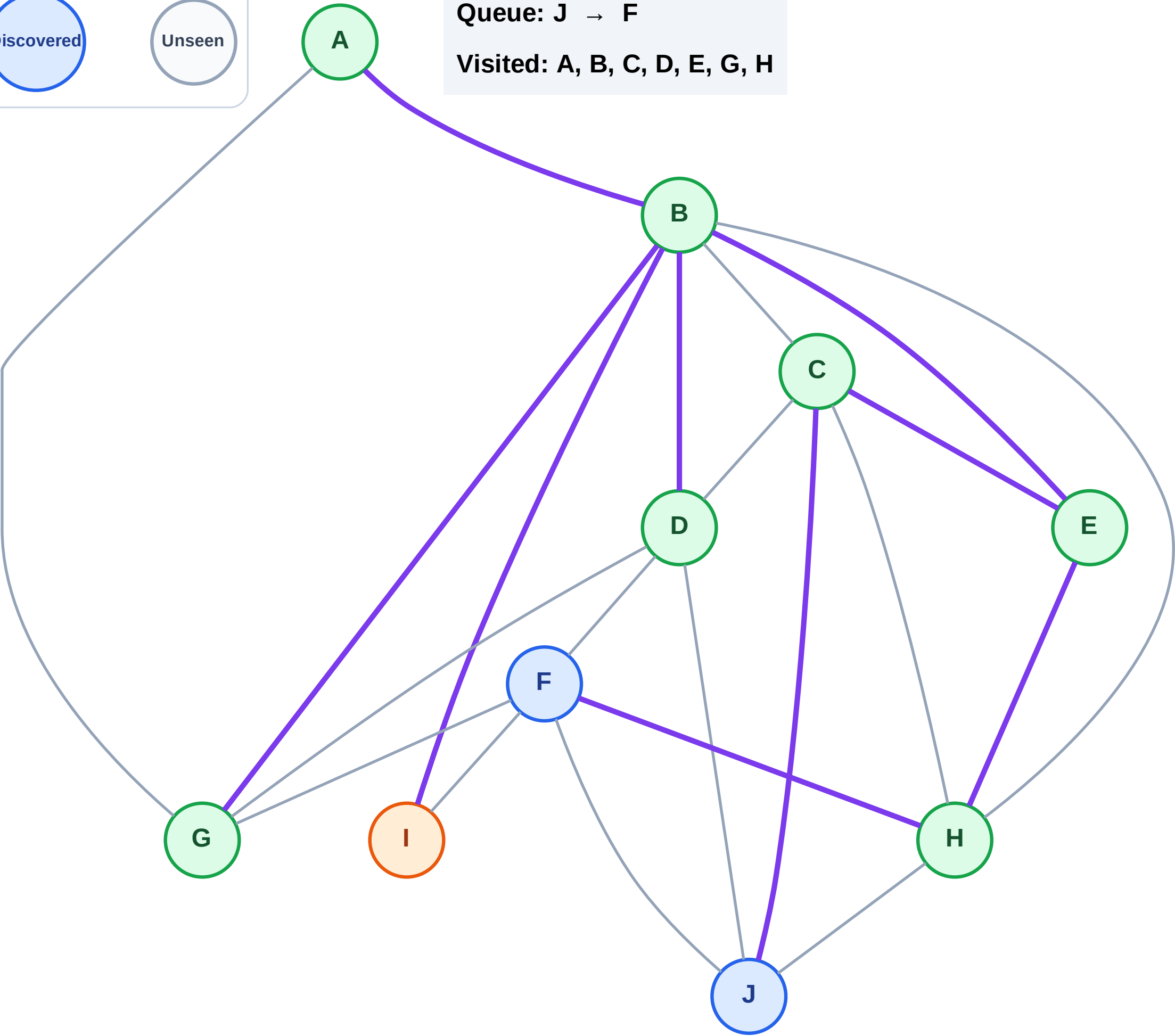
Processing

Discovered

Unseen

Queue: J → F

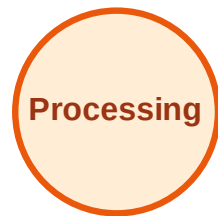
Visited: A, B, C, D, E, G, H



BFS Traversal

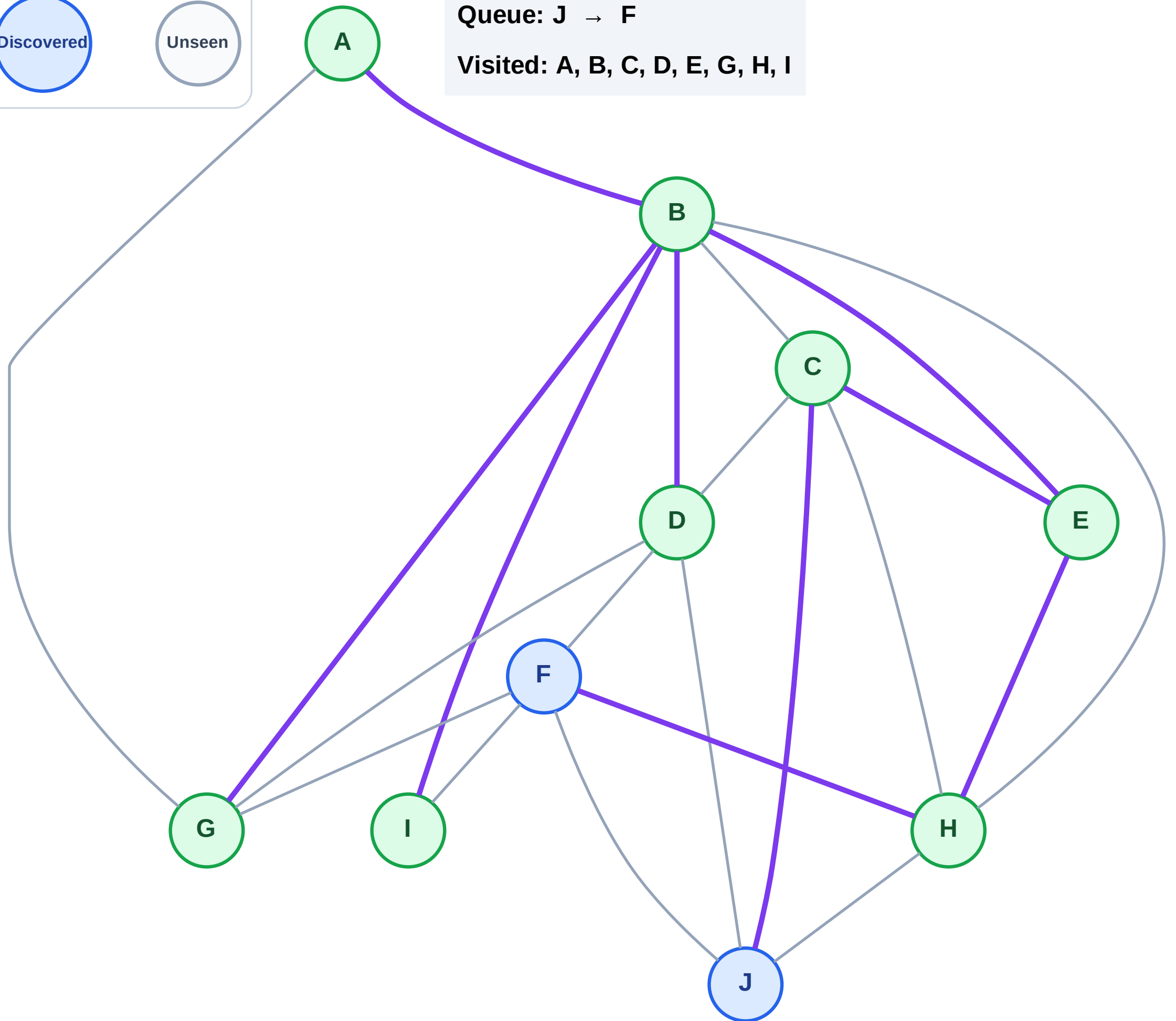
Step 17: No new neighbors from I

Legend



Queue: J → F

Visited: A, B, C, D, E, G, H, I



BFS Traversal

Step 18: Process node J

Legend

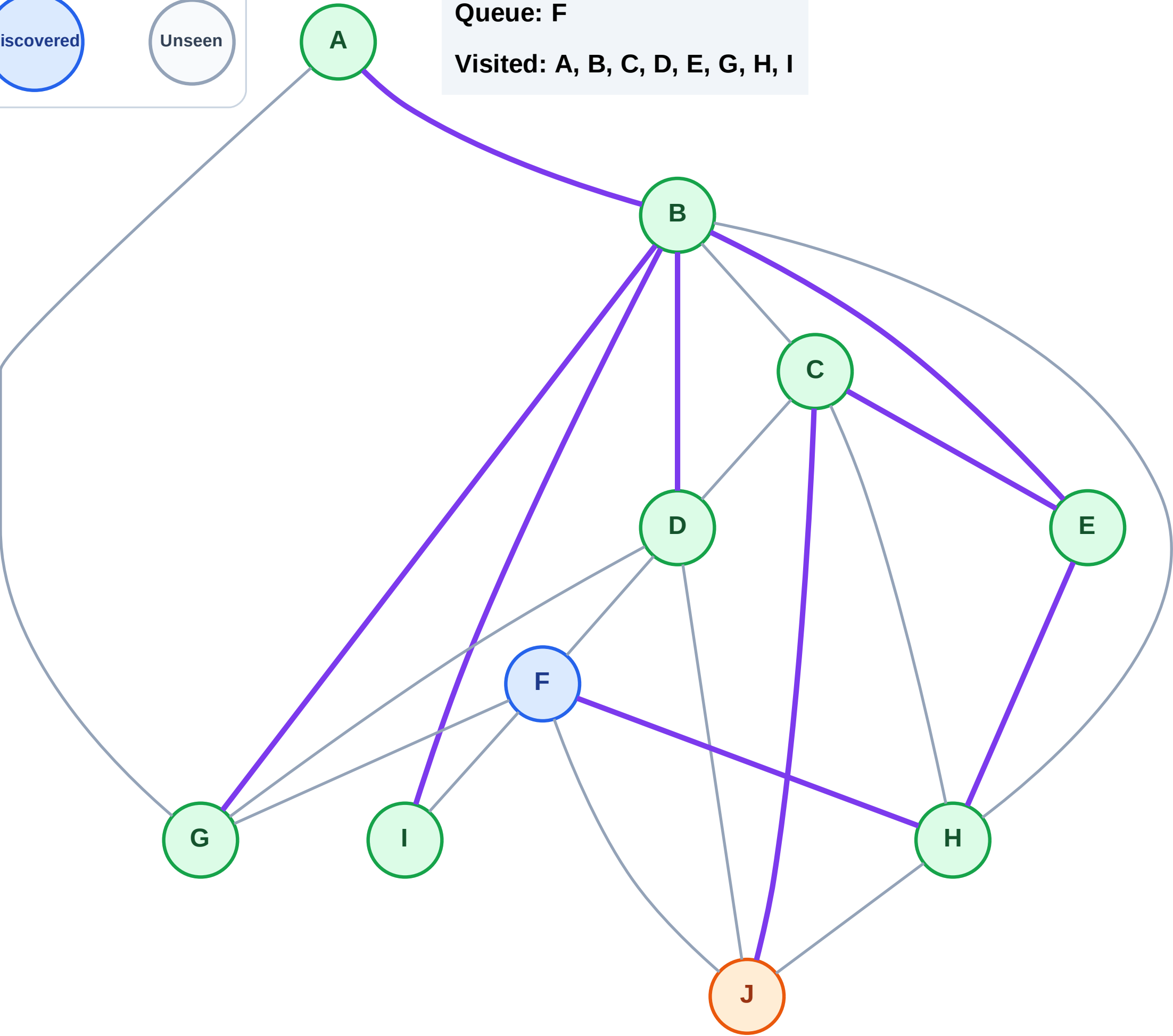
Complete

Processing

Discovered

Unseen

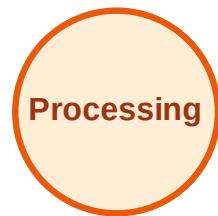
Queue: F
Visited: A, B, C, D, E, G, H, I



BFS Traversal

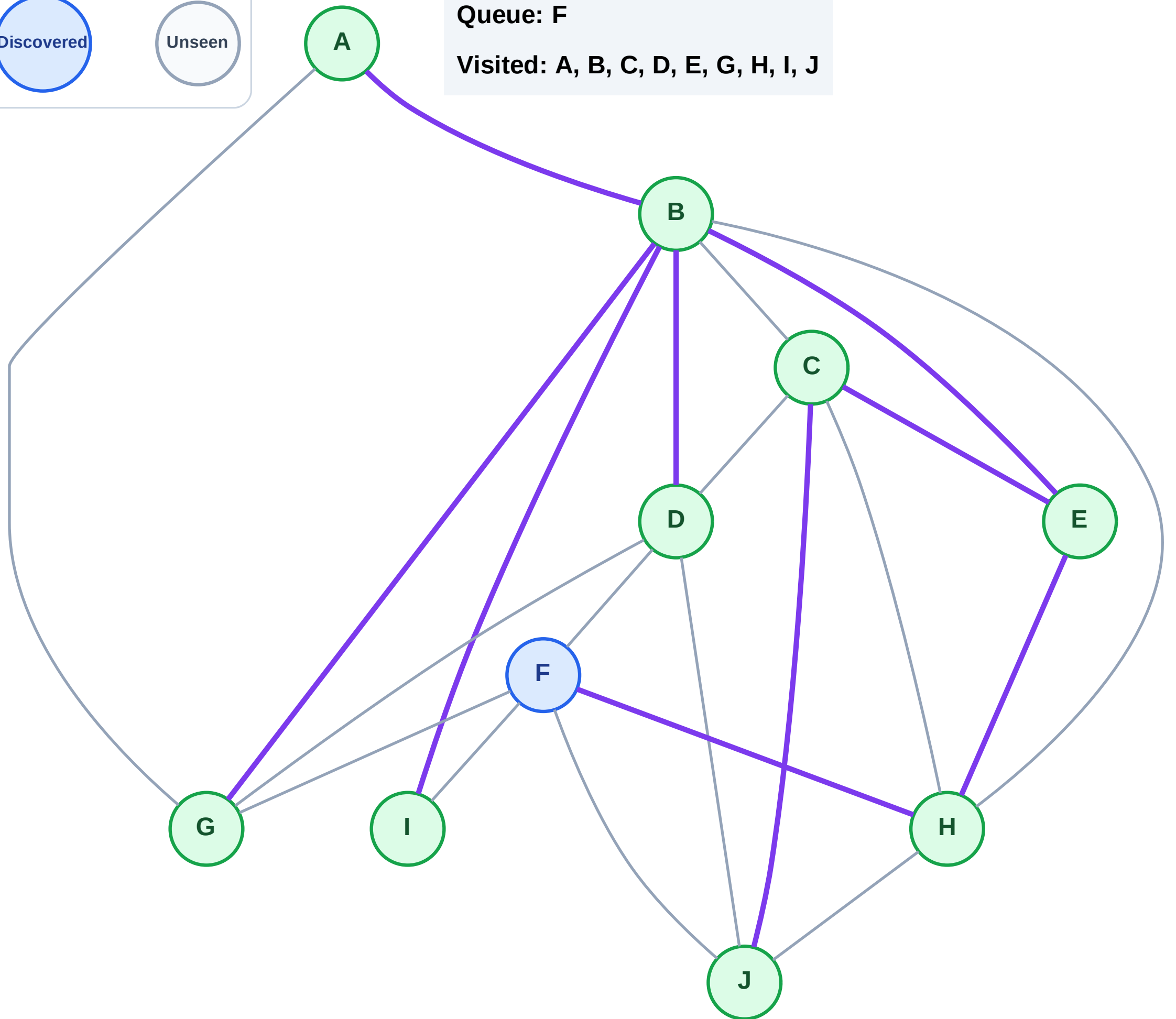
Step 19: No new neighbors from J

Legend



Queue: F

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

Step 20: Process node F

Legend

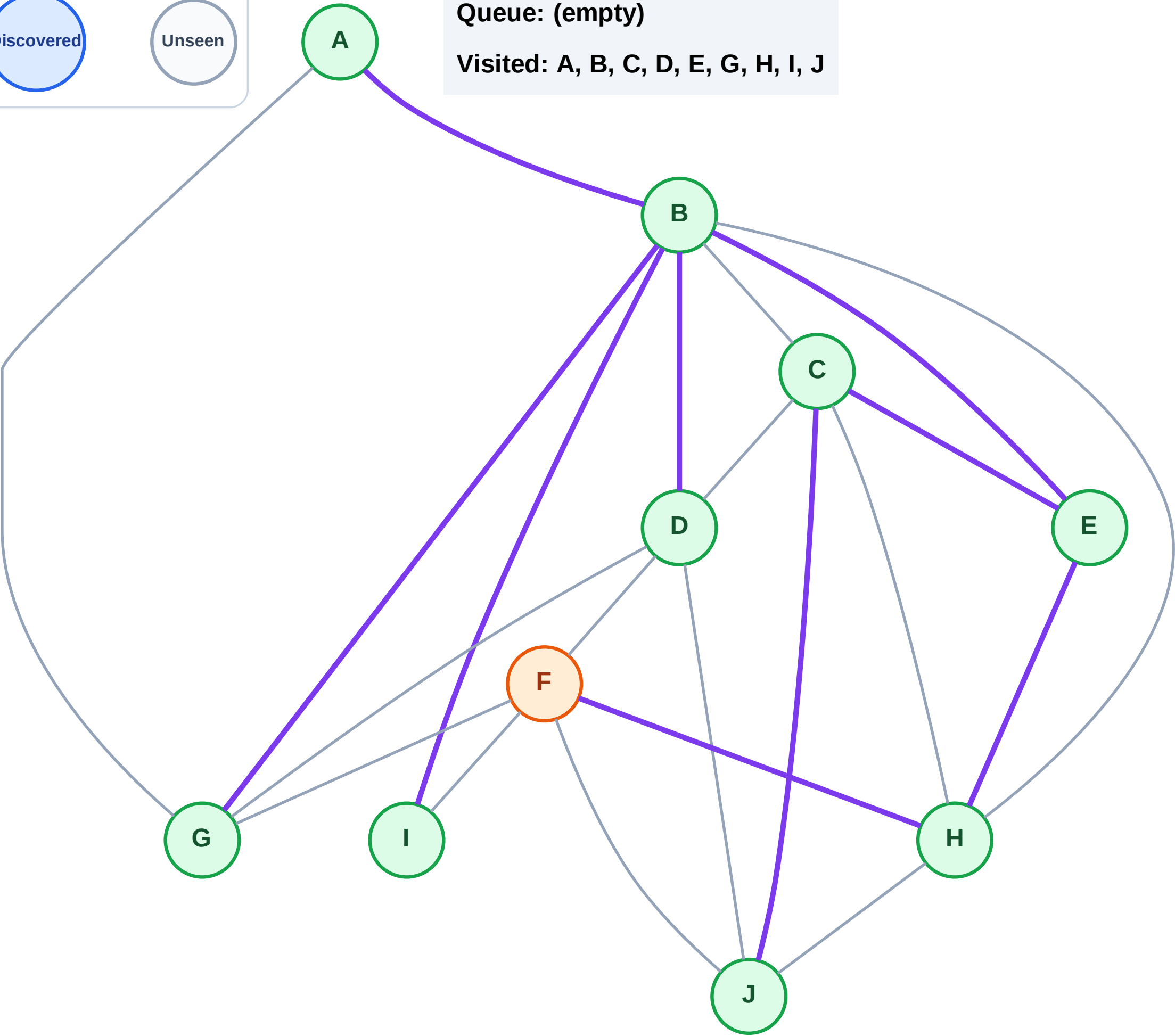
Complete

Processing

Discovered

Unseen

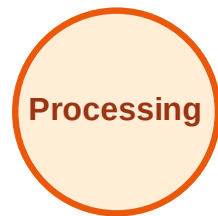
Queue: (empty)
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

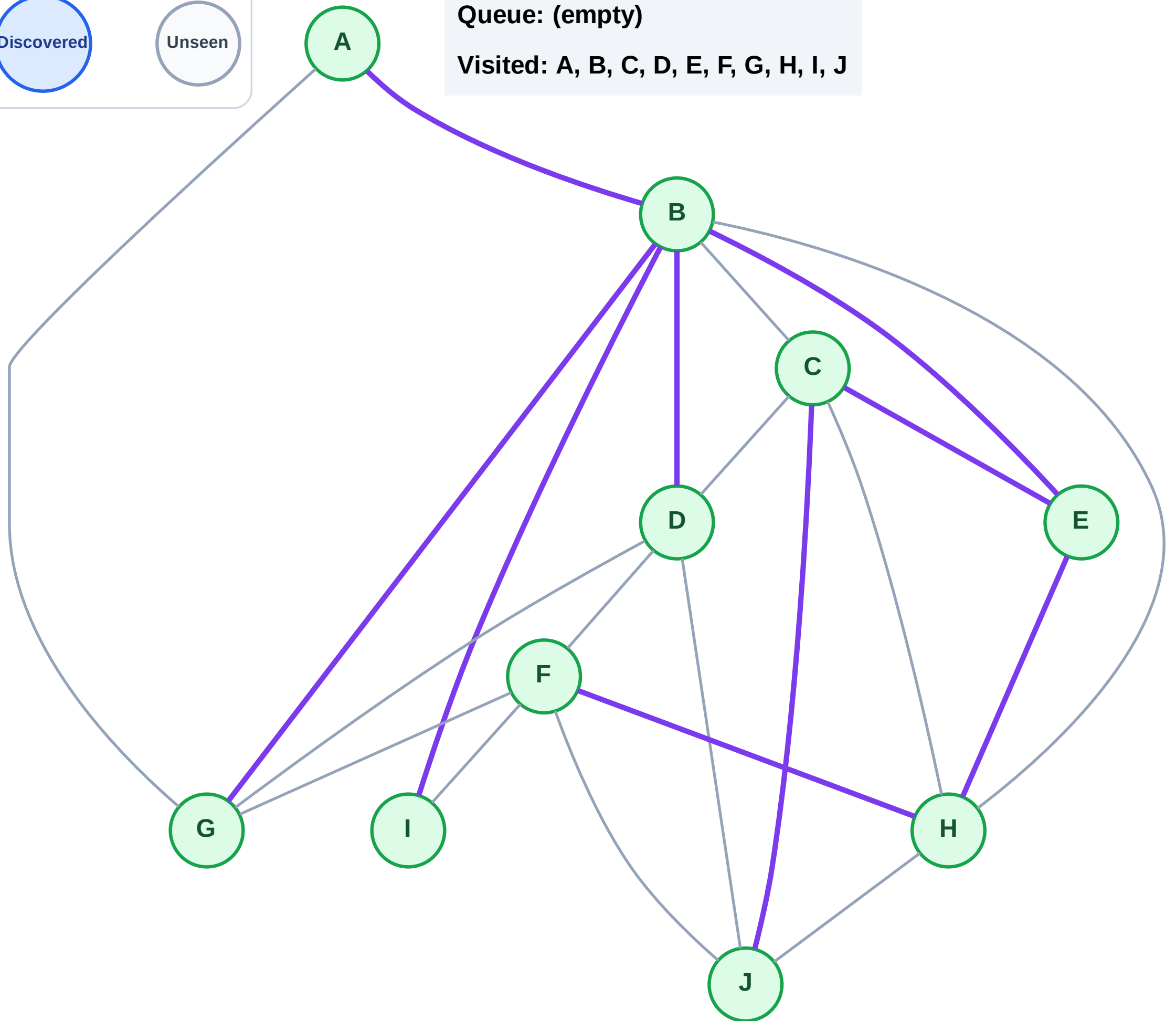
Step 21: No new neighbors from F

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

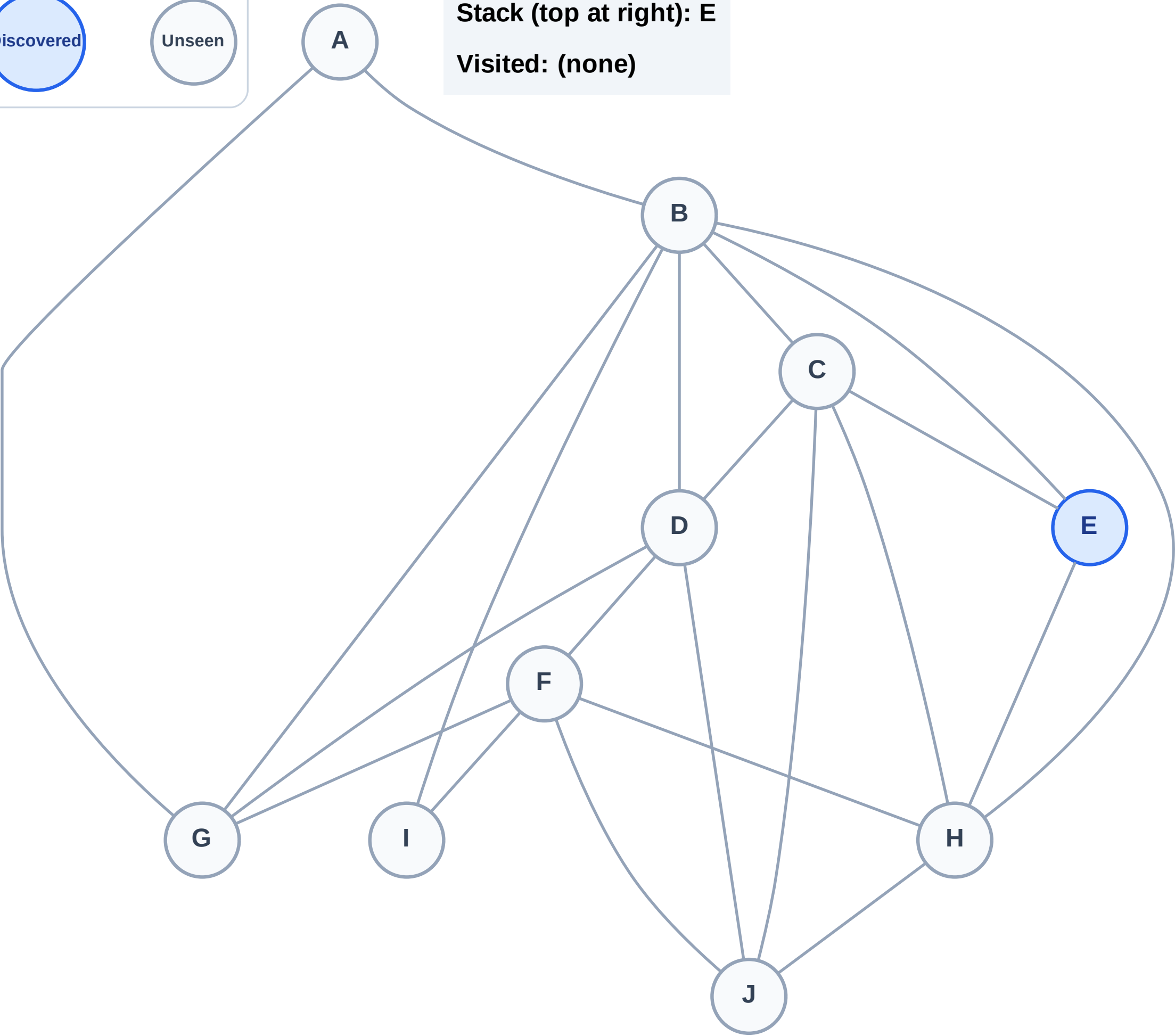
Complete

Processing

Discovered

Unseen

Stack (top at right): E
Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

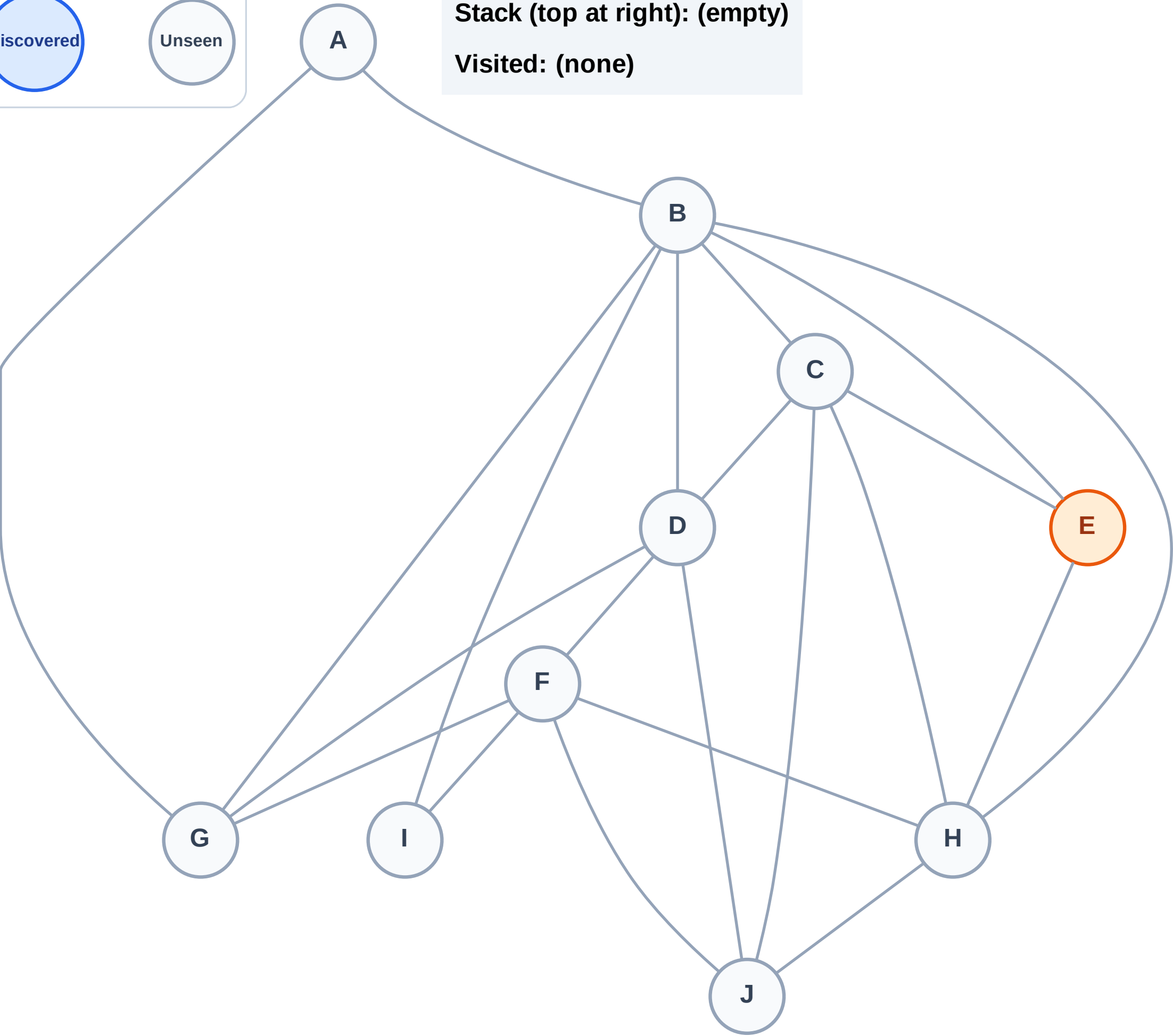
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)

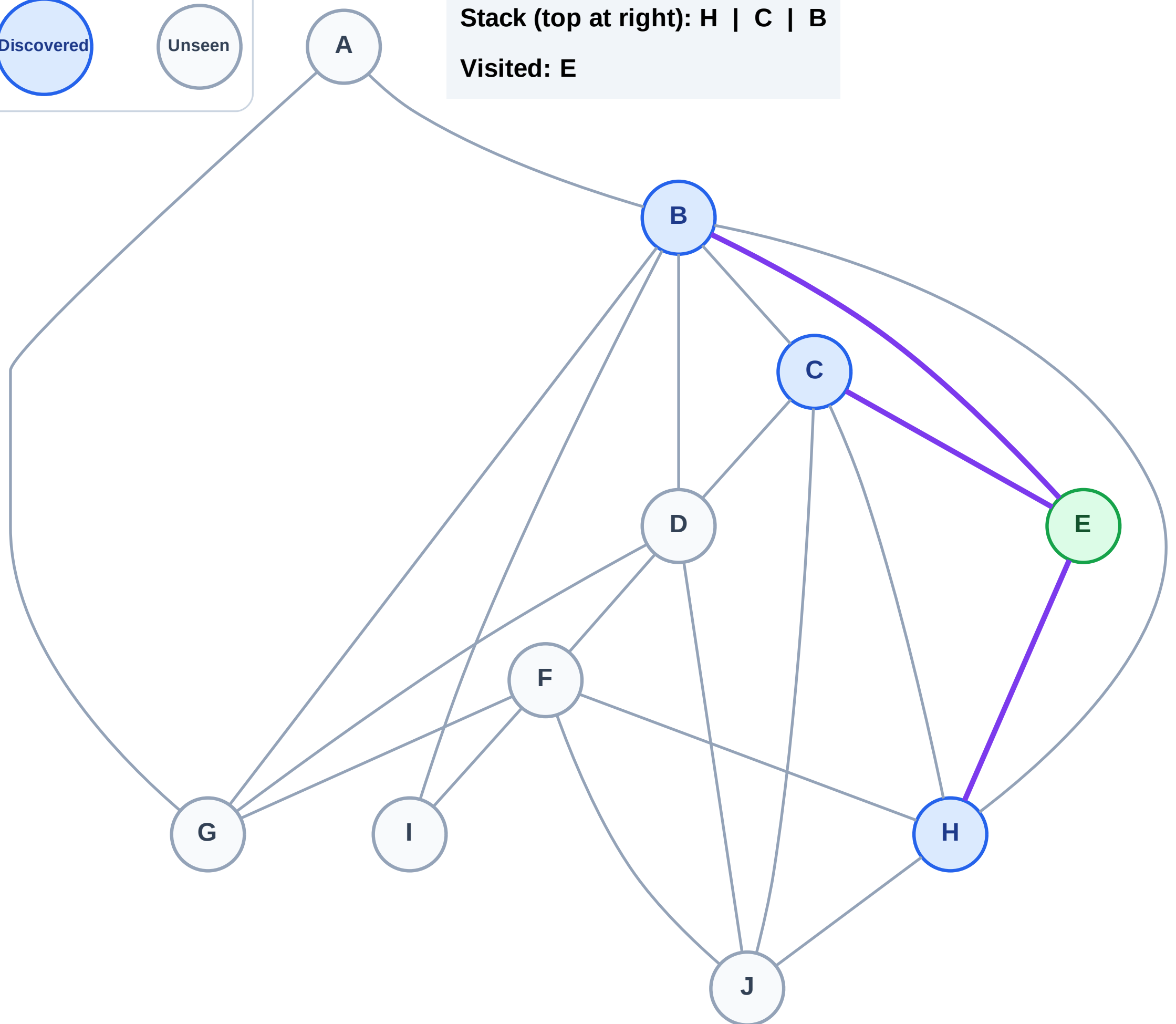


Step 3: Discover H, C, B from E

Stack (top at right): H | C | B

Legend

Unseen



DFS Traversal

Step 4: Process node B

Legend

Complete

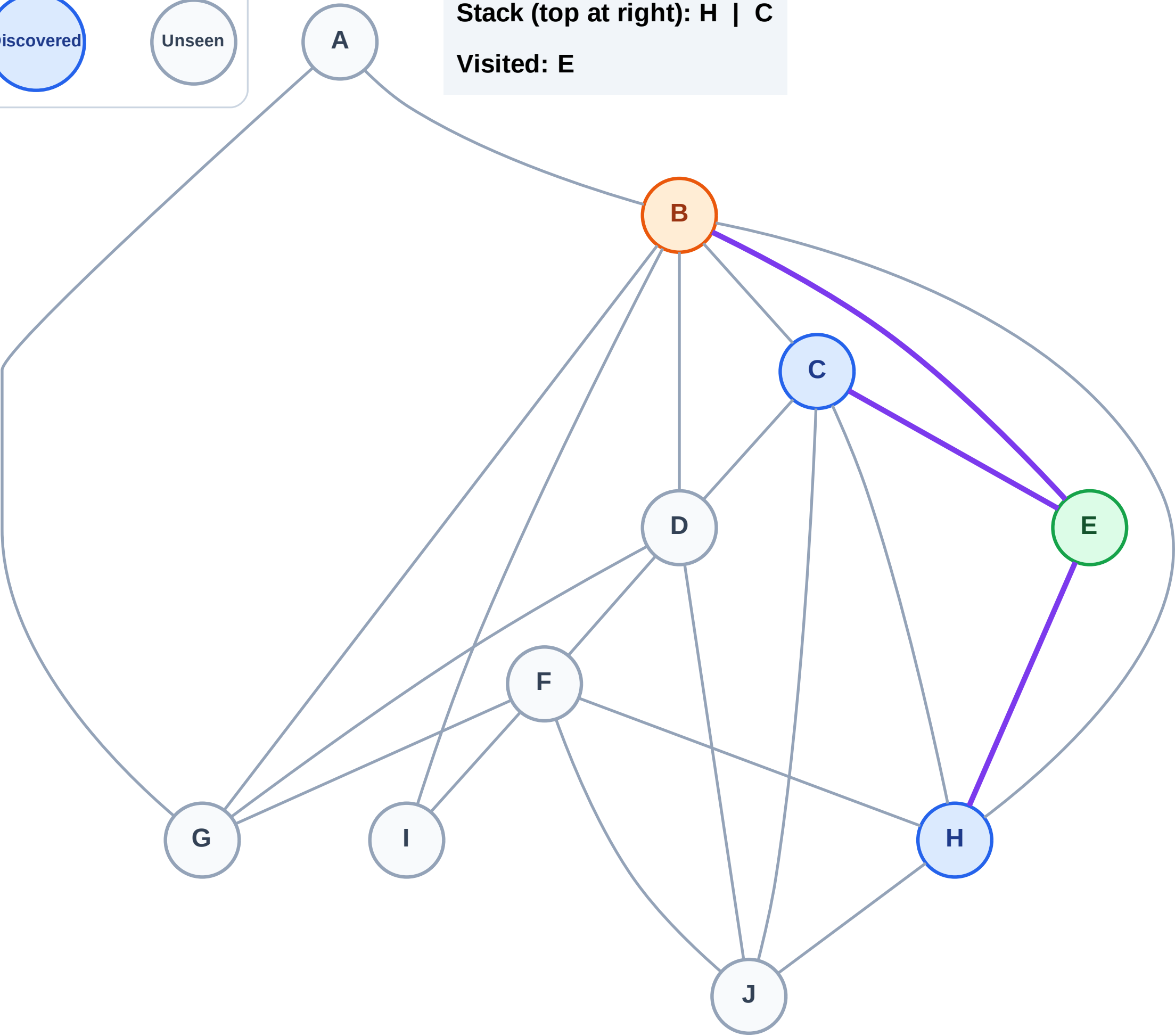
Processing

Discovered

Unseen

Stack (top at right): H | C

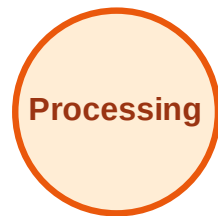
Visited: E



DFS Traversal

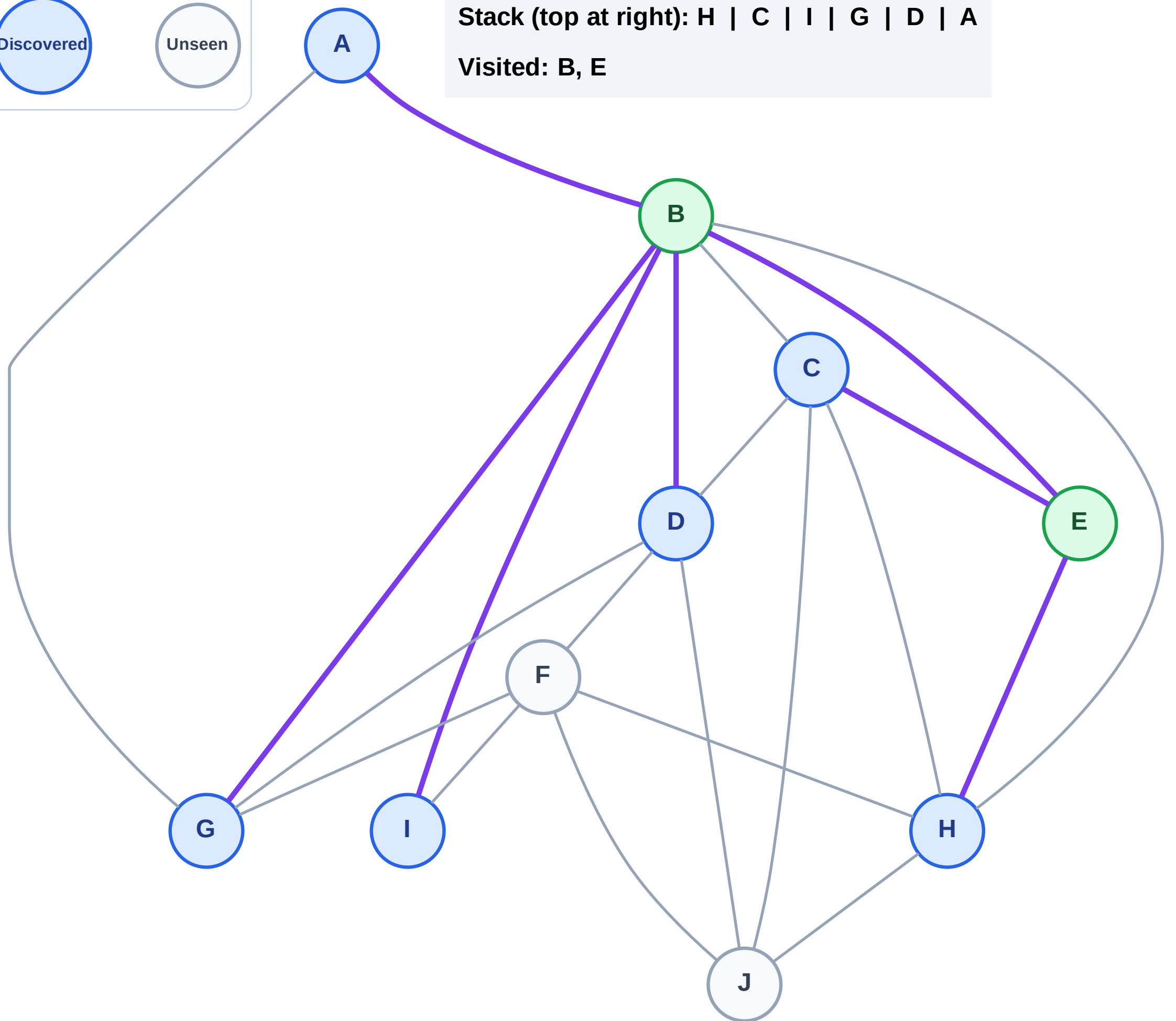
Step 5: Discover I, G, D, A from B

Legend



Stack (top at right): H | C | I | G | D | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

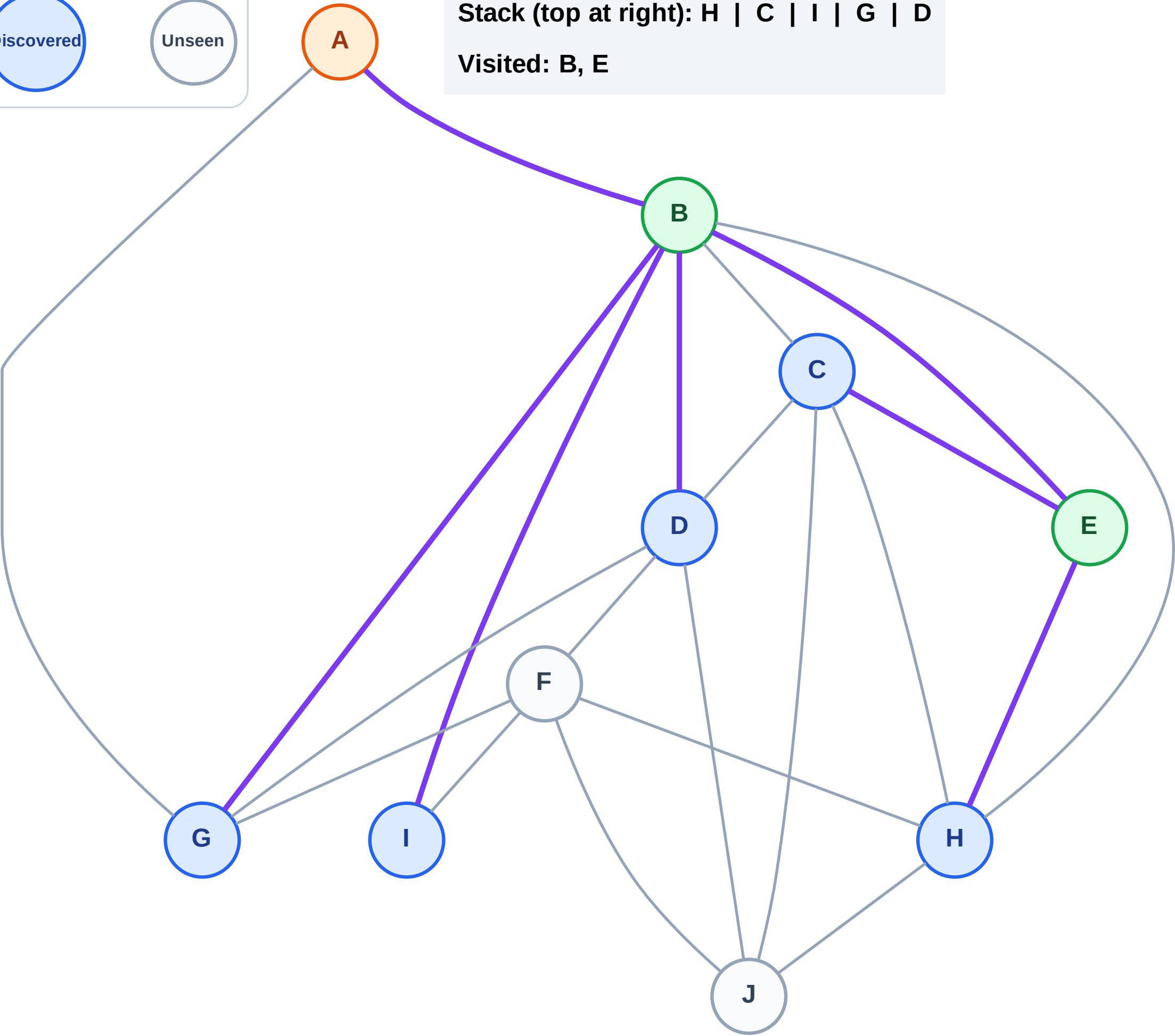
Processing

Discovered

Unseen

Stack (top at right): H | C | I | G | D

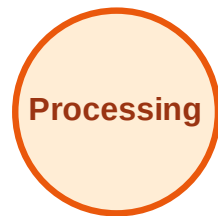
Visited: B, E



DFS Traversal

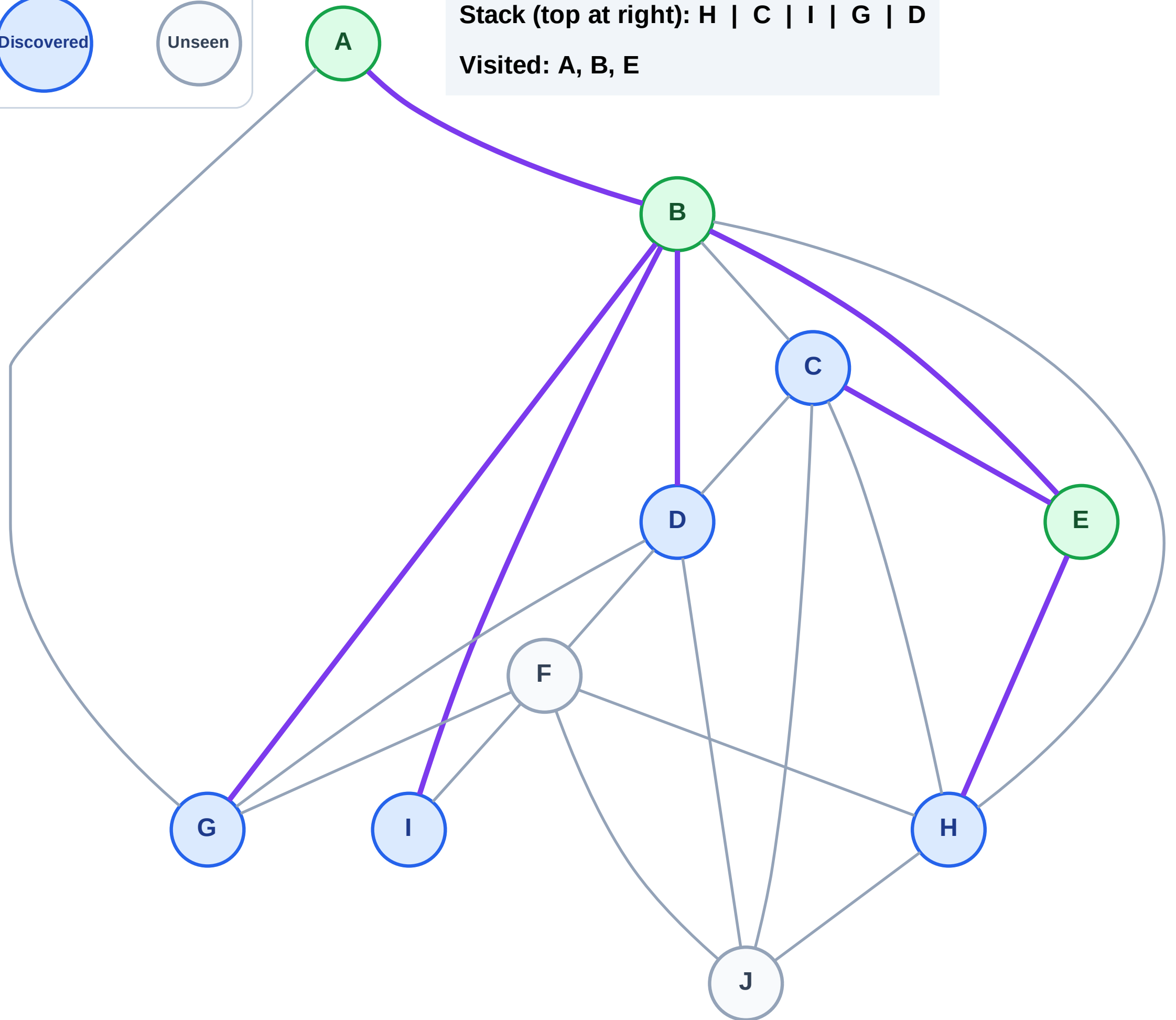
Step 7: No new neighbors from A

Legend



Stack (top at right): H | C | I | G | D

Visited: A, B, E



DFS Traversal

Step 8: Process node D

Legend

Complete

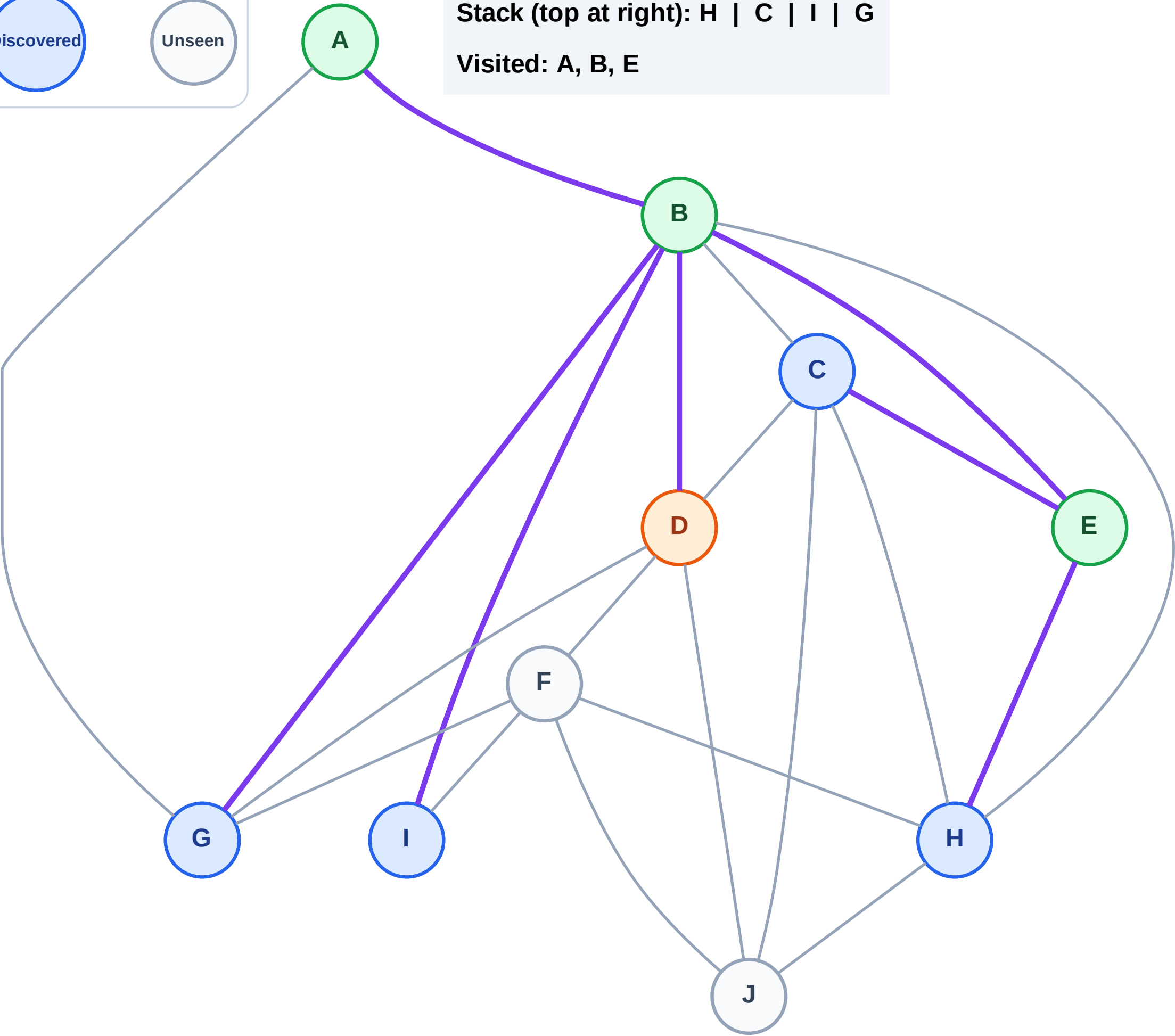
Processing

Discovered

Unseen

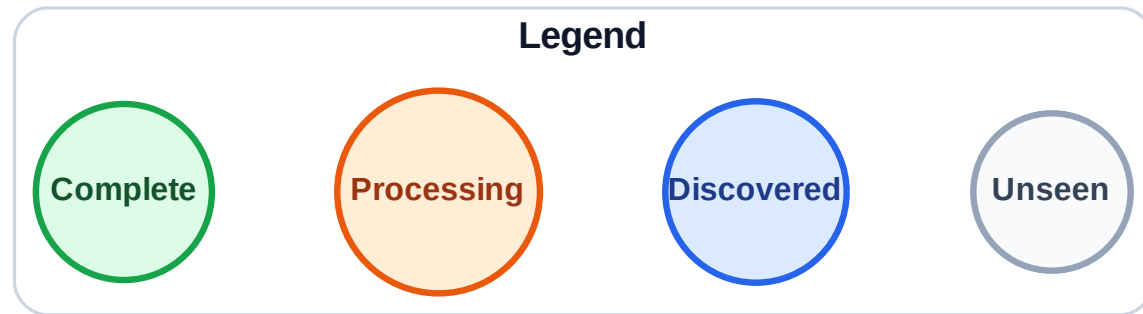
Stack (top at right): H | C | I | G

Visited: A, B, E

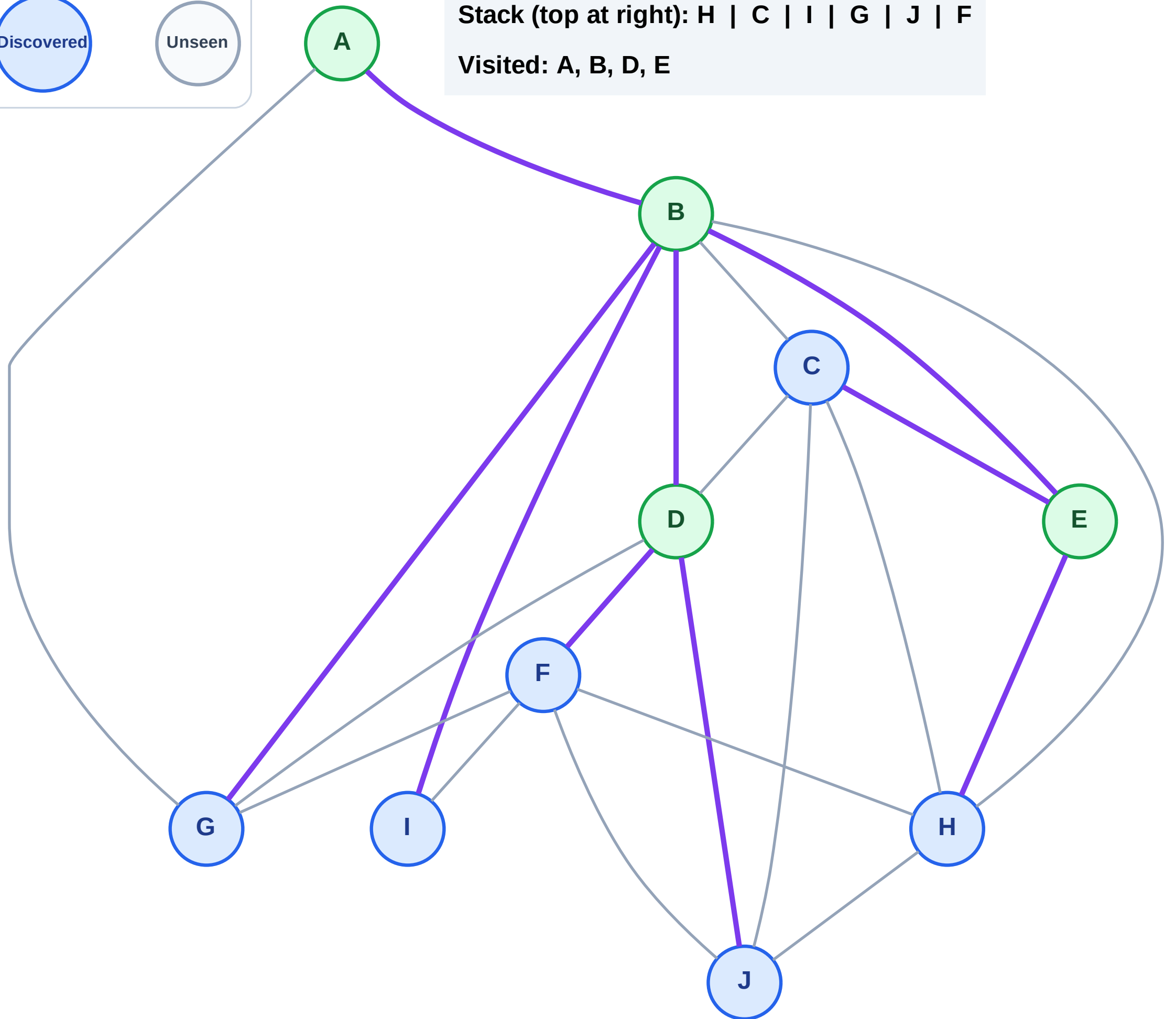


DFS Traversal

Step 9: Discover J, F from D



Stack (top at right): H | C | I | G | J | F
Visited: A, B, D, E



DFS Traversal

Step 10: Process node F

Legend

Complete

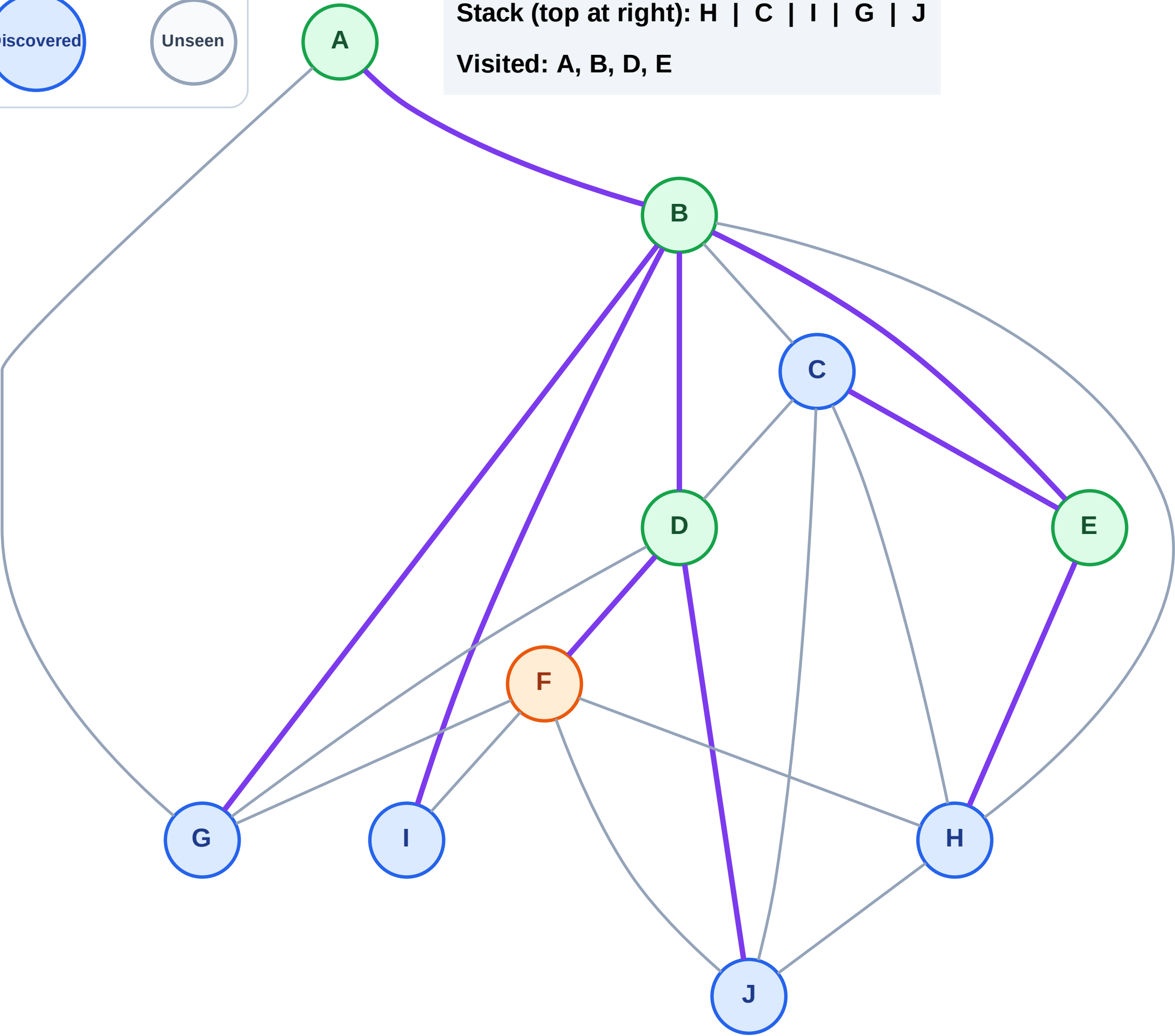
Processing

Discovered

Unseen

Stack (top at right): H | C | I | G | J

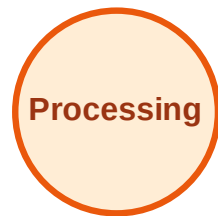
Visited: A, B, D, E



DFS Traversal

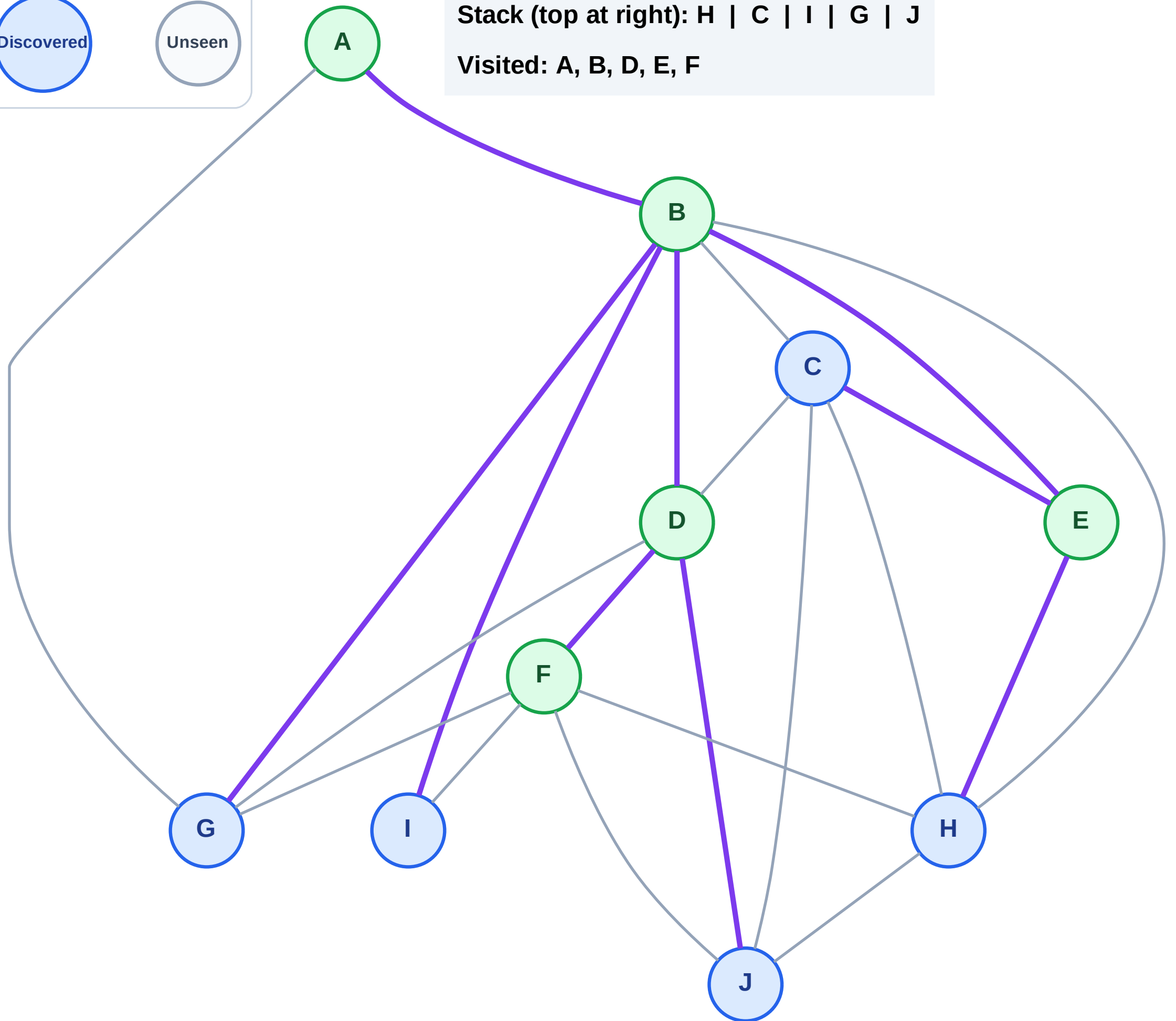
Step 11: No new neighbors from F

Legend



Stack (top at right): H | C | I | G | J

Visited: A, B, D, E, F



DFS Traversal

Step 12: Process node J

Legend

Complete

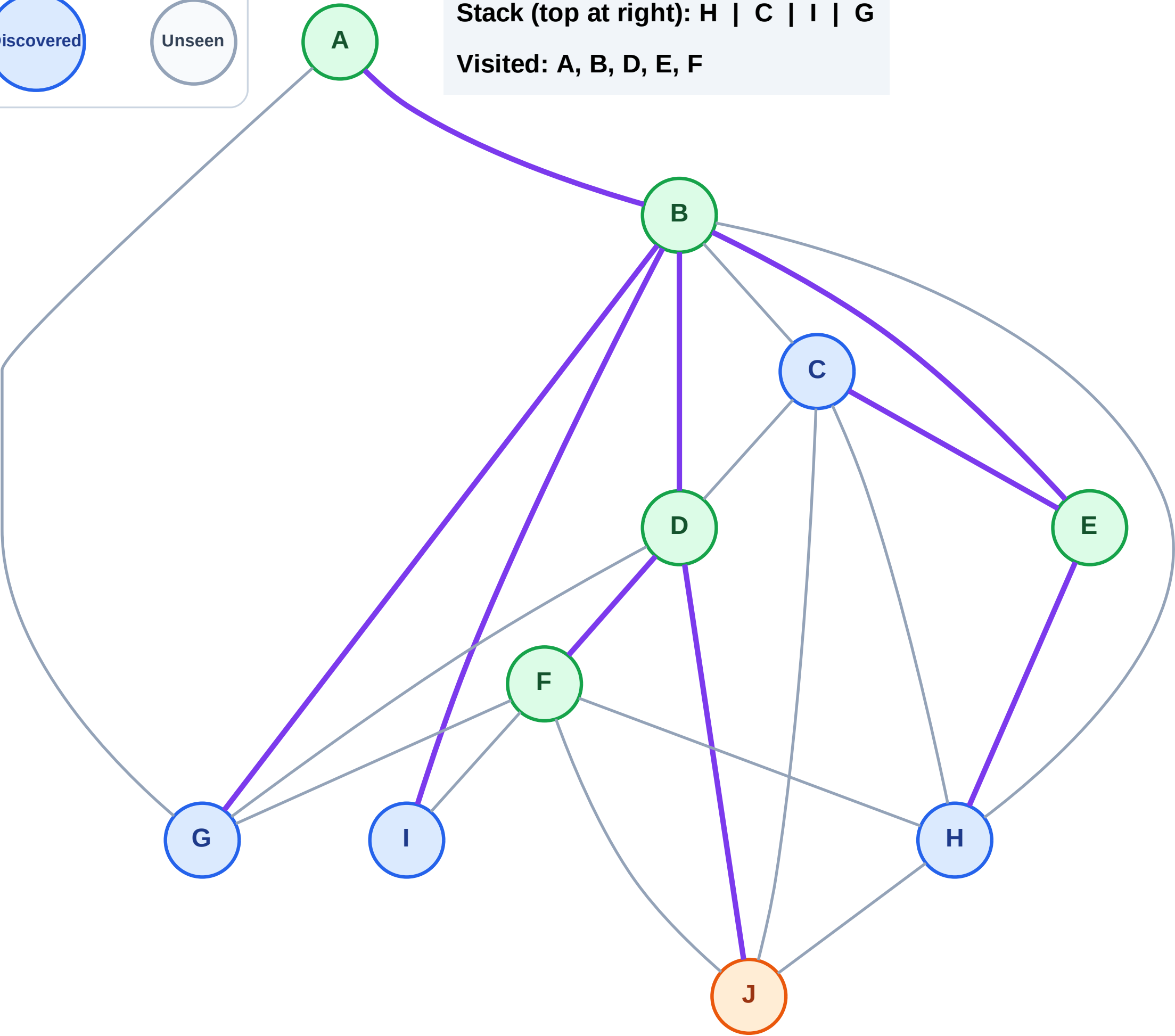
Processing

Discovered

Unseen

Stack (top at right): H | C | I | G

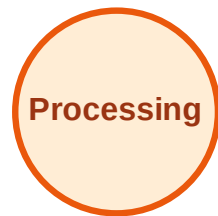
Visited: A, B, D, E, F



DFS Traversal

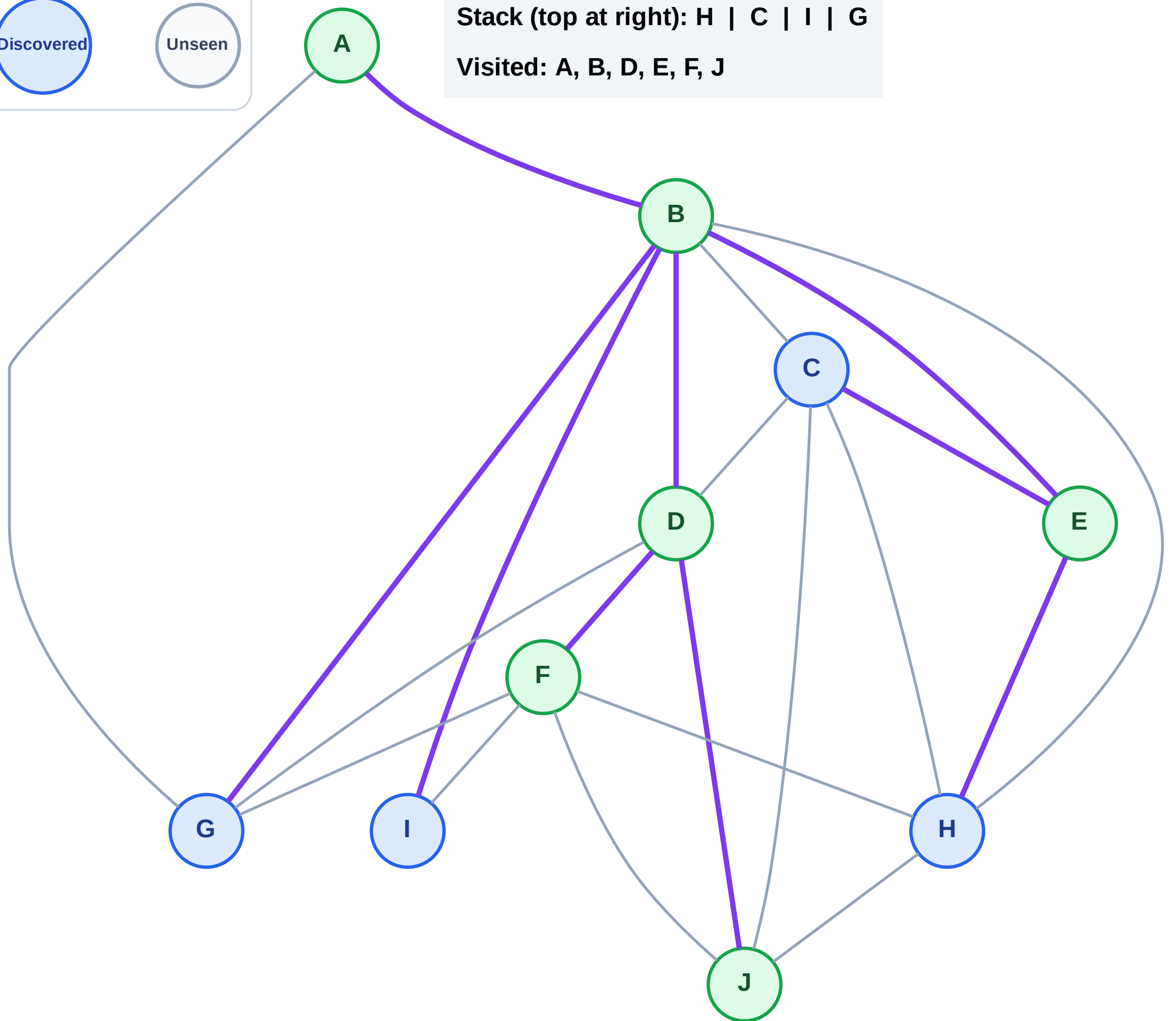
Step 13: No new neighbors from J

Legend



Stack (top at right): H | C | I | G

Visited: A, B, D, E, F, J



DFS Traversal

Step 14: Process node G

Legend

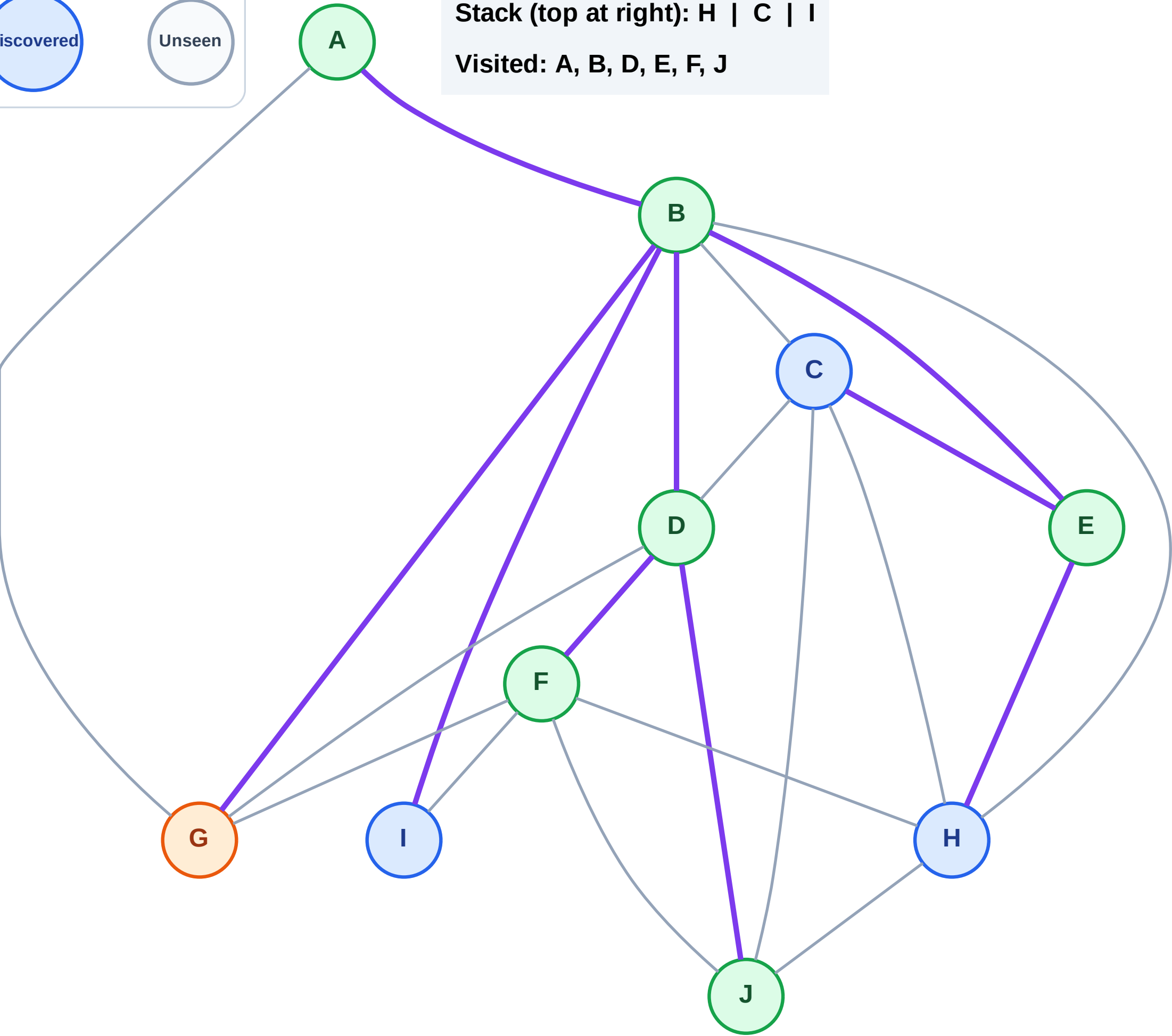
Complete

Processing

Discovered

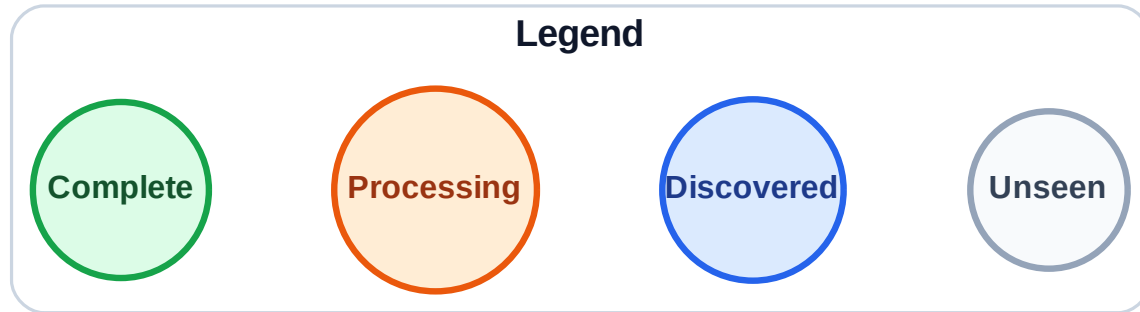
Unseen

Stack (top at right): H | C | I
Visited: A, B, D, E, F, J



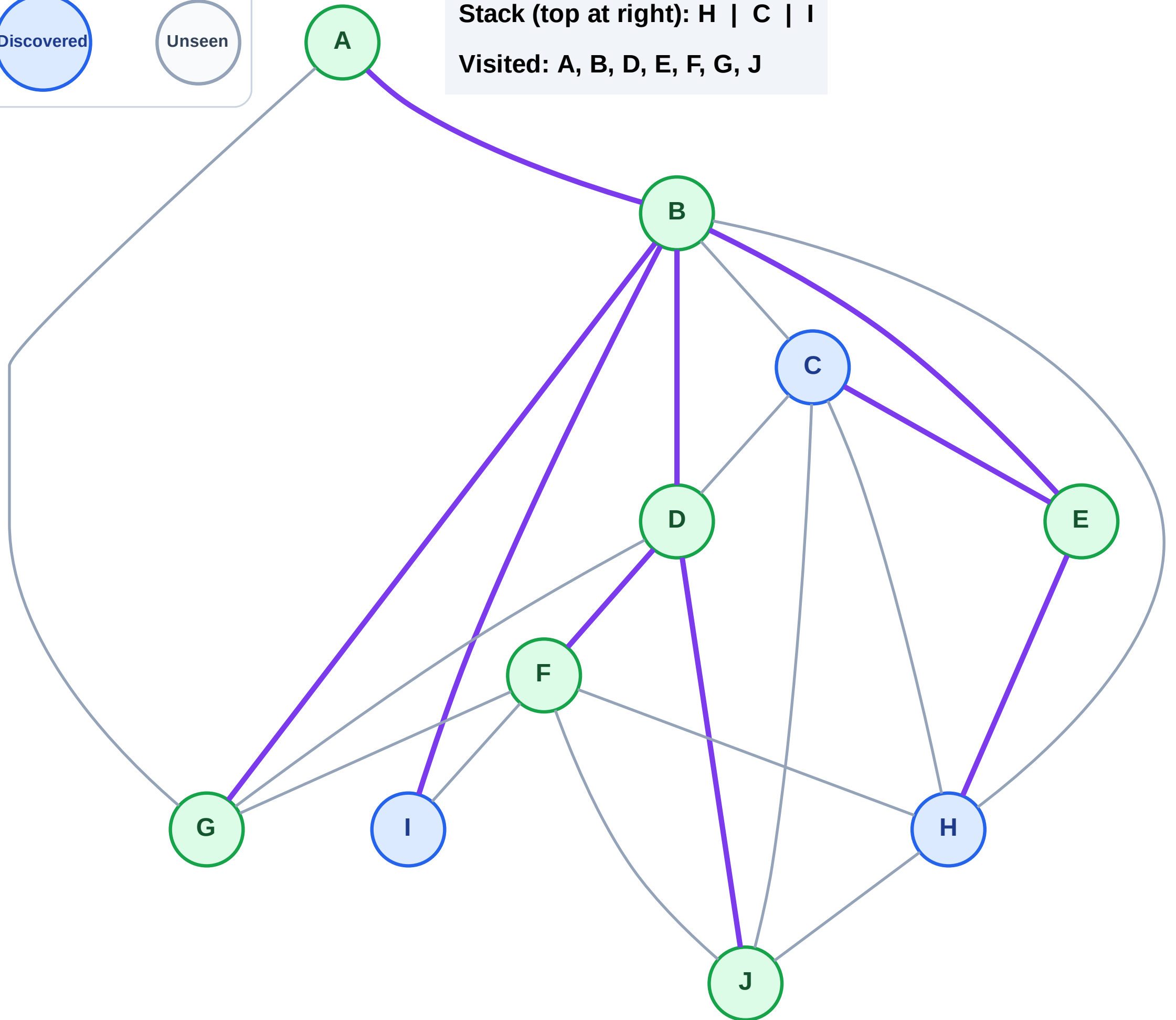
Step 15: No new neighbors from G

Step 15: No new neighbors from G



Stack (top at right): H | C | I

Visited: A, B, D, E, F, G, J



DFS Traversal

Step 16: Process node I

Legend

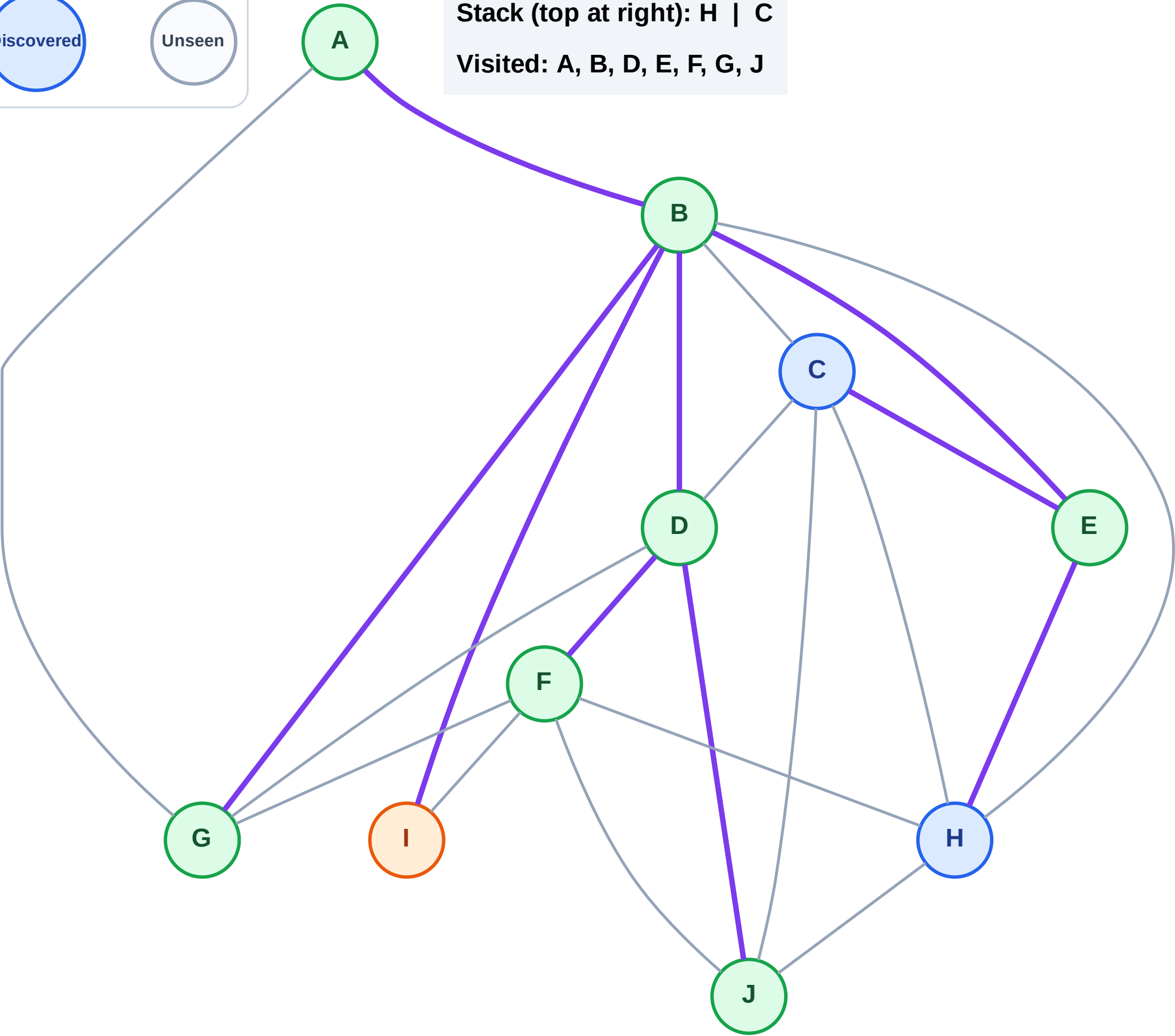
Complete

Processing

Discovered

Unseen

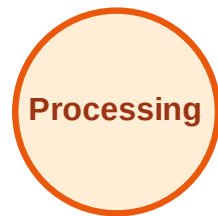
Stack (top at right): H | C
Visited: A, B, D, E, F, G, J



DFS Traversal

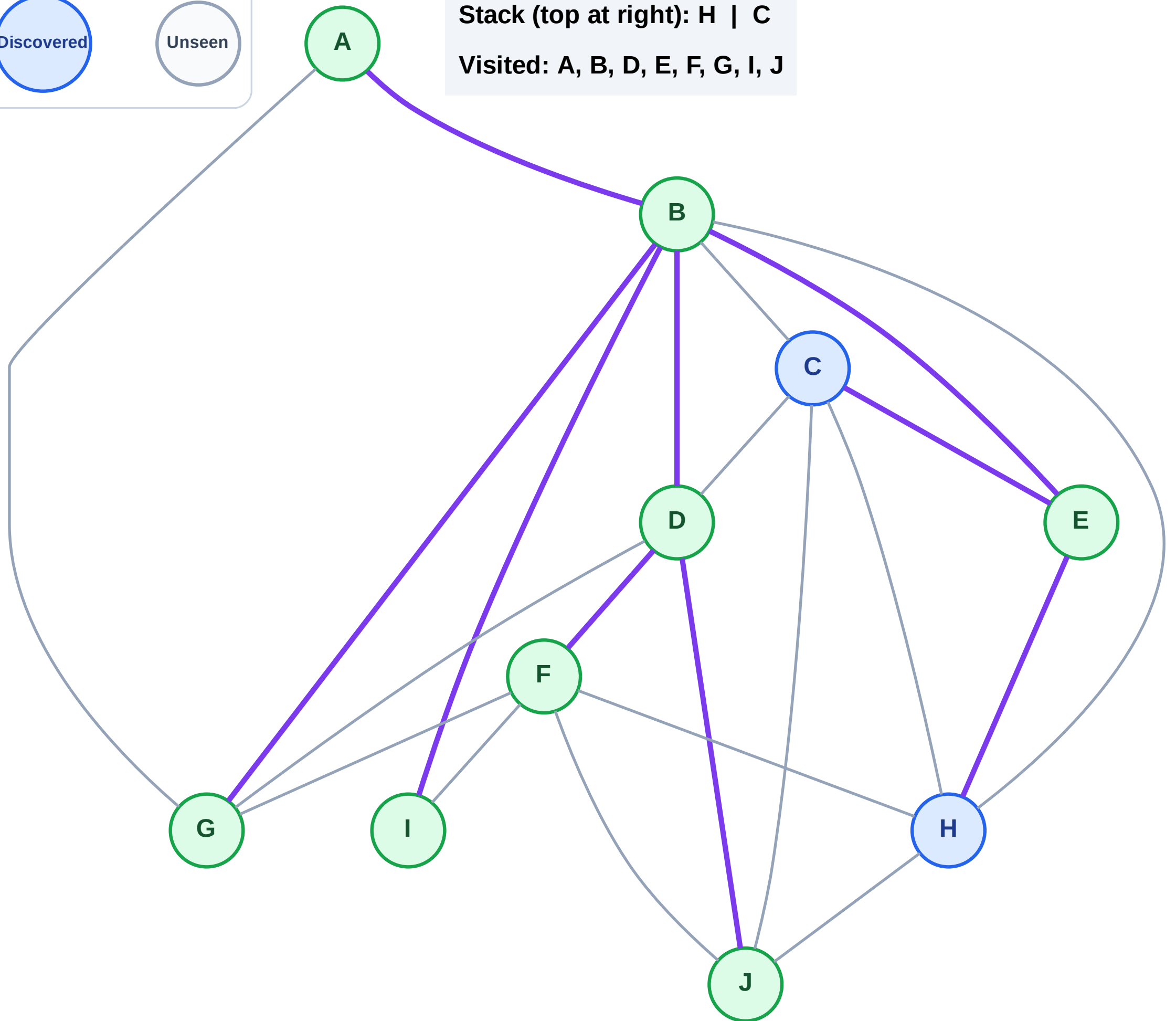
Step 17: No new neighbors from I

Legend



Stack (top at right): H | C

Visited: A, B, D, E, F, G, I, J



DFS Traversal

Step 18: Process node C

Legend

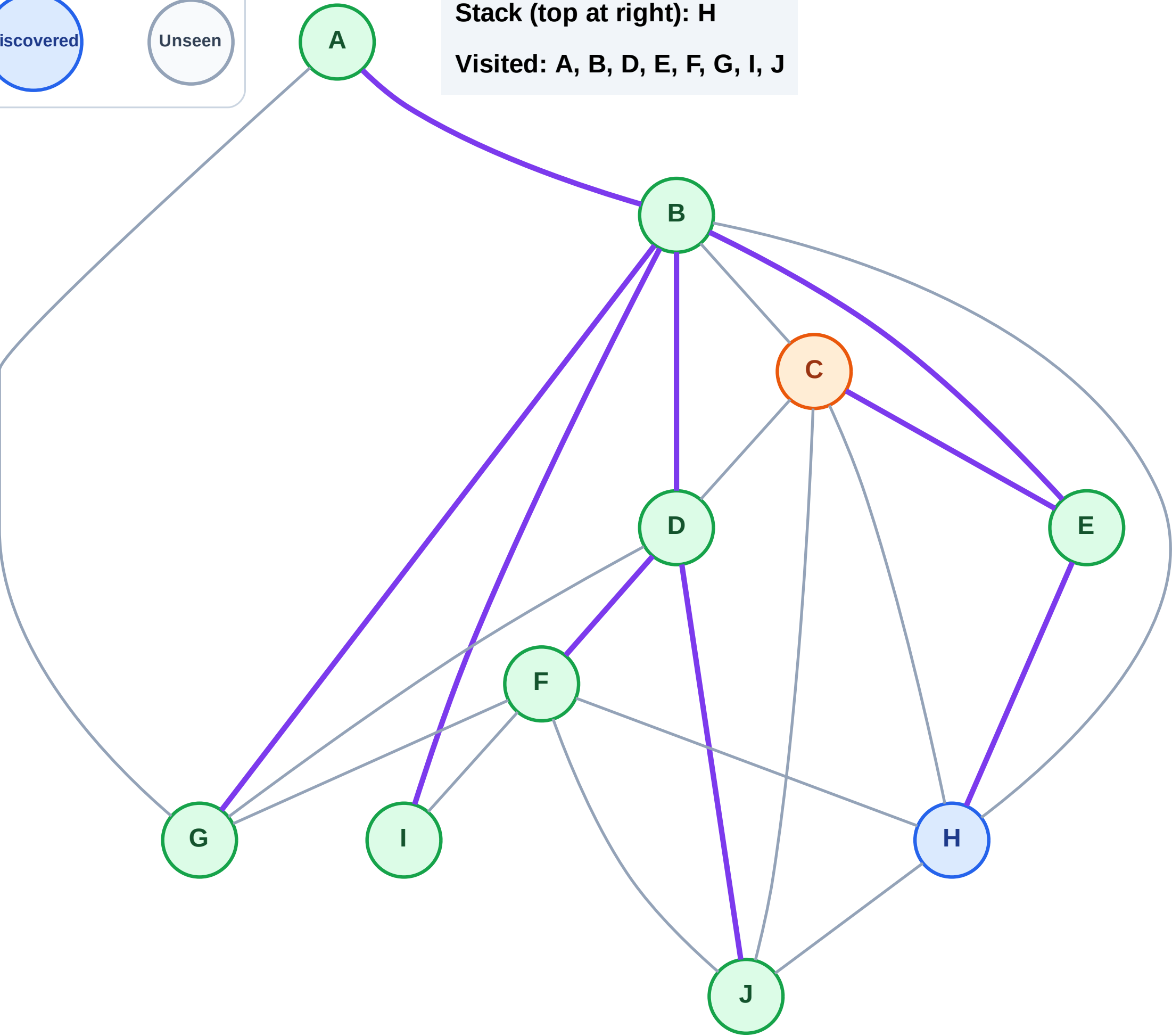
Complete

Processing

Discovered

Unseen

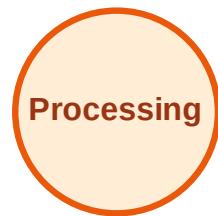
Stack (top at right): H
Visited: A, B, D, E, F, G, I, J



DFS Traversal

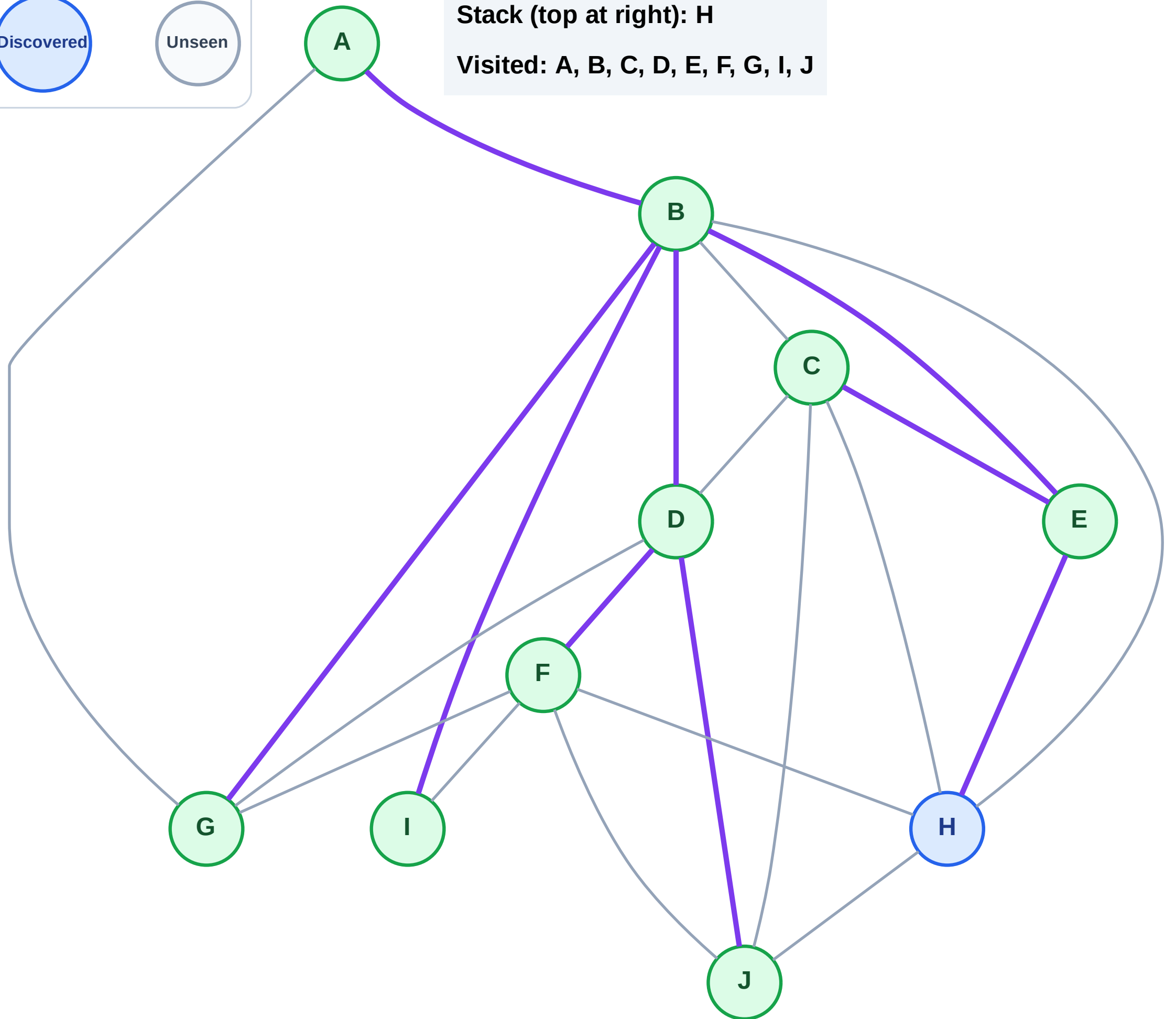
Step 19: No new neighbors from C

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 20: Process node H

Legend

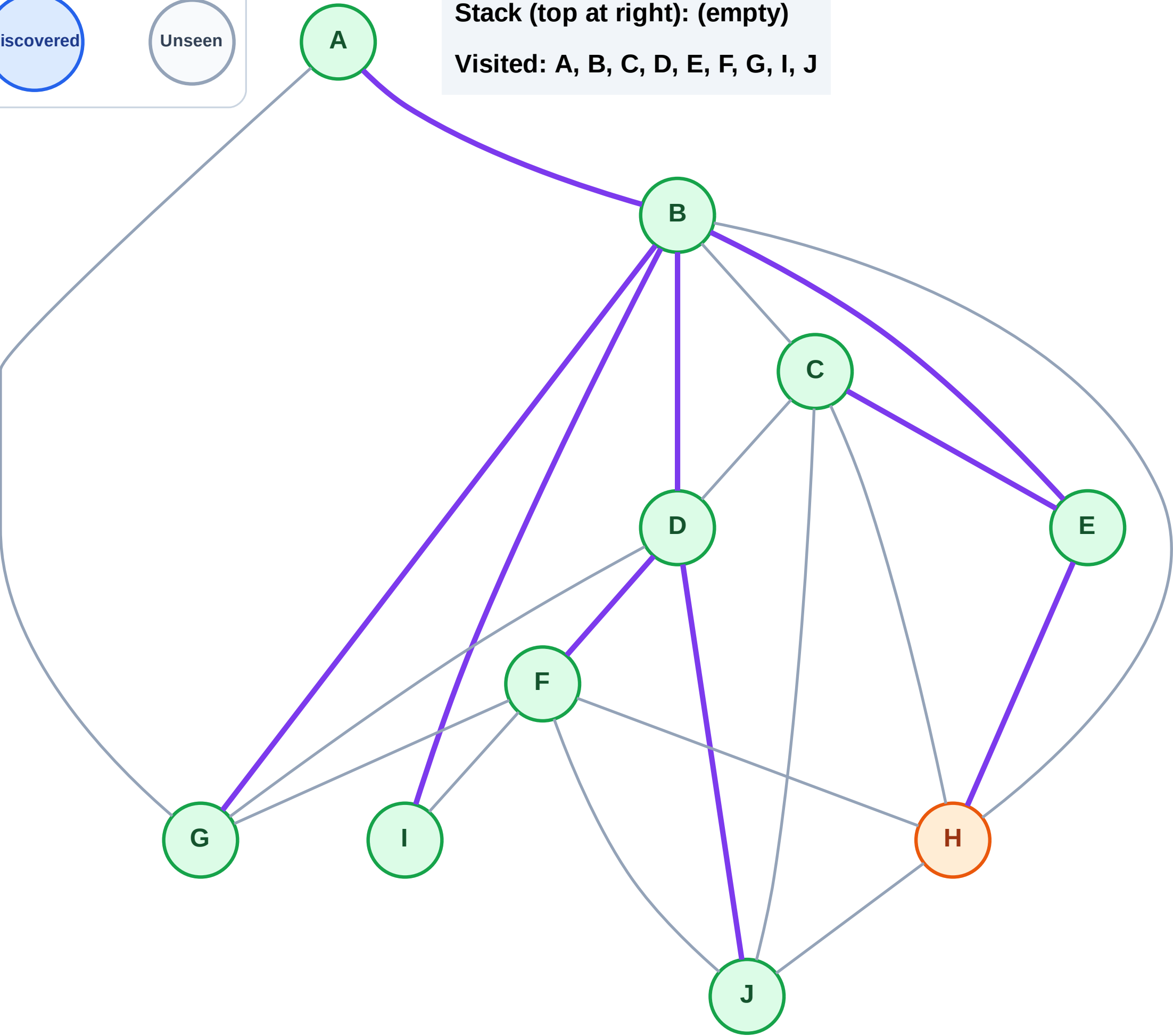
Complete

Processing

Discovered

Unseen

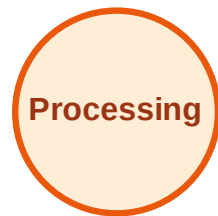
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

